

Session 6.1

From Chat Window to Agentic AI: What an Agent Is, and Where It Pays

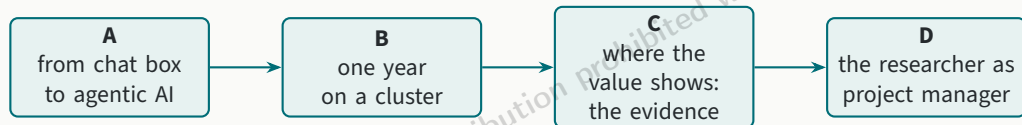
Part one of five: chatbot, copilot, agent, agent team — what makes AI agentic, and one year of an agent on a cluster, in the instructor's own projects



Zhigang Feng

The Question, and the Route

What is an agent, what makes AI agentic —
and what did it change for one researcher?



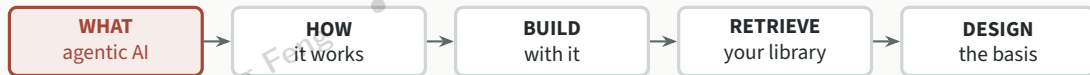
- ▶ Session 5 ended with a language model you can score. Everything it produced was **advice**: you still ran the code, moved the files and checked the numbers.
- ▶ This session closes the gap between advice and action — and opens with a year in which it closed on one account.
- ▶ **Four parts: what an agent is, what it did on a cluster, what 637 scientists report, and what the job becomes.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §1; Codreanu et al. (2026).

Session 6.1

From Chat Box to Agentic AI

What makes AI agentic, and what does not change



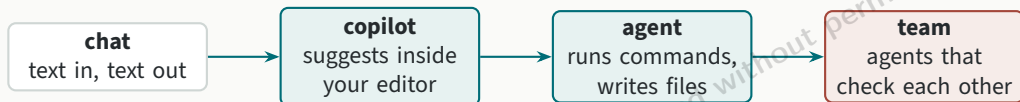
From a Chat Box to Agent Teams, 2022–2026: The Rows Deck 1.3 Left Out



- ▶ Since Deck 1.3 drew this: models that think before they answer (OpenAI o1-preview, Sep 2024); the protocol moved to a Linux Foundation body (Dec 2025); task length doubles about every 89 days (METR, Jan 2026).
- ▶ **Each step made the work cheaper; none made checking it cheaper.**

Source: Deck 1.3's figure; rows added from Anthropic Economic Index (28 Apr 2025), Linux Foundation (9 Dec 2025), METR Time Horizon 1.1 (29 Jan 2026), Lu et al. (2026, Nature), Codreanu et al. (2026), all read 23 Sep 2026; OpenAI (12 Sep 2024).

Chat, Copilot, Agent: A Spectrum, Not a Switch



what changes across the strip is not the model — it is how much of your machine it can touch

- ▶ One measured point on the strip: in April 2025, 79% of conversations with a terminal agent were *automation* (the model does the task) against 49% in the chat window — the same index Deck 1.2 used to place AI inside occupations.
- ▶ A chat window is still the right tool for one question and one answer.
- ▶ **You choose a position on this strip for each task, and you can move left when a task turns out to be delicate.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.1–§3.2; Anthropic Economic Index, “AI’s impact on software development”, 28 Apr 2025 (read 23 Sep 2026).

What Makes AI Agentic: Four Marks, All Needed

tool use

calls Python, R, Stata, the shell

autonomy

a goal, not a prompt: it plans the steps

iteration

reads the error, fixes, reruns

persistence

conventions and plans kept on disk

An **agent** is a model in a loop, given tools and pointed at a goal: the cleverness is in the model, the *capability* in the loop.

- ▶ The marks are cumulative. Tool use without autonomy is a console; autonomy without iteration dies at the first error; without persistence nothing survives the session.
- ▶ **“Agentic” is a degree, not a type: how much of the loop runs without you.**

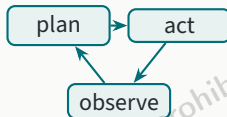
Source: Miao & Feng, *AI for Economists*, ch. 15, §3.1.

Agent, Workflow, Agent Team: Who Decides the Next Step

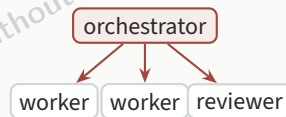
a workflow: code decides



an agent: the model decides



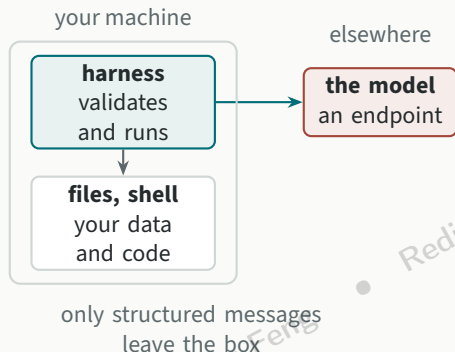
a team: agents split the work



- ▶ In a workflow, models and tools are “orchestrated through predefined code paths”; in an agent, models “dynamically direct their own processes and tool usage”.
- ▶ A workflow is cheap and predictable; an agent handles surprises; a team buys breadth and pays in coordination.
- ▶ The advice is to start at the left: “finding the simplest solution possible, and only increasing complexity when needed”.
- ▶ **Capability and risk both sit with whoever chooses the next step.**

Source: Anthropic, “Building effective agents”, 19 Dec 2024 (read 24 Sep 2026); Miao & Feng, *AI for Economists*, ch. 15, §3.1 and §4.3.

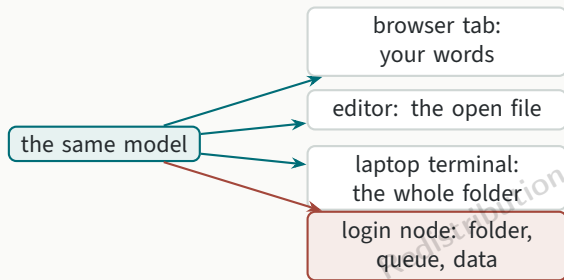
Three Parts, and Only One of Them Is Clever



- ▶ **The harness:** an ordinary program on your computer. It reads and writes files, runs commands, and enforces the rules you set.
- ▶ **The model:** not on your computer. An endpoint that reads messages and names the next call. That is the whole of it.
- ▶ **Files and tools:** your data, your code, your shell — all local.
- ▶ **The intelligence sits in one part; the discipline sits in the other two.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.2.

Same Model, Different Reach: Where the Loop Runs

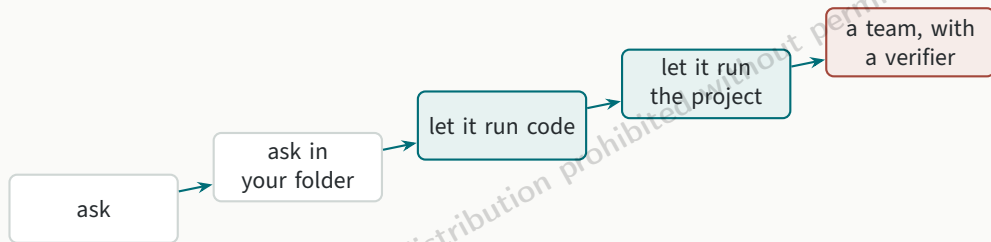


The capability that matters for research is not the model's reasoning. It is the *blast radius* of what it is allowed to reach — and the fourth box is Part B's whole story.

- ▶ The reach dial is where the harness runs, not which model answers: on a login node it reaches the folder, the scheduler, and data that never has to move.
- ▶ **“Install the AI on the server” was not a new capability. It was the same loop, moved to where the files are.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.2; this deck's Part B, by role.

The Ladder: From a Chat Window to a Verified Team



climb one rung only when
the rung below is boring

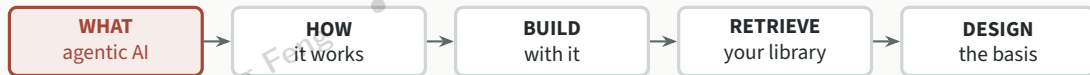
- ▶ **Every rung up buys throughput and costs verification. The last rung is the only one where an agent checks an agent.**
- ▶ The year on the cluster reached the fourth rung; Deck 6.3 builds the fifth.

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.3.

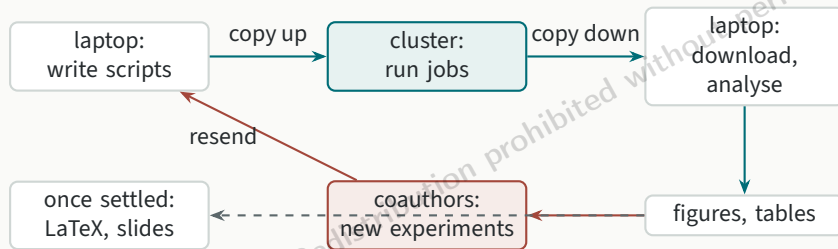
Session 6.1

One Year on a Cluster

The loop before, the loop after, and what the agent does now



The Pre-AI Loop: Write Locally, Run Remotely, Analyse Locally

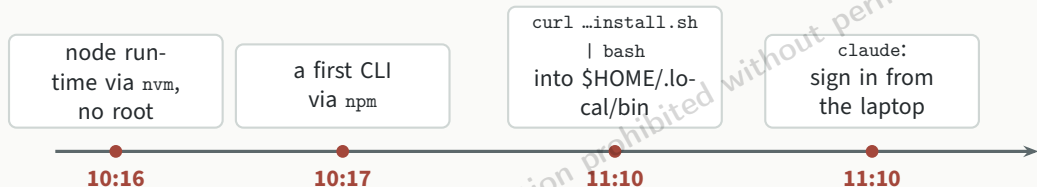


every arrow was a manual transfer; analysis and writing never touched the cluster; the loop closed weekly at best

- **Two machines, one person carrying files between them. The cluster computed; everything else happened where the researcher sat.**

Source: Instructor's project record (the national-debt pipeline), 2025–26, by role.

Install Day: Three Commands on a Login Node, 20 Apr 2026

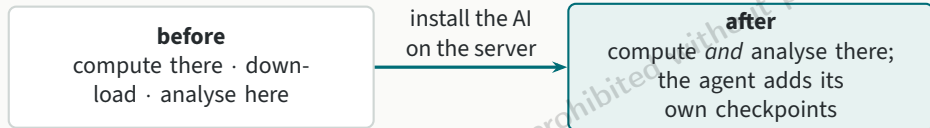


the harness's own state file records the first start at 11:10:22 that morning; by 23 Sep 2026 it counted 224 startups across 24 project folders

- ▶ No administrator was involved: a user-local runtime, a user-local binary, a browser sign-in on the laptop.
- ▶ **The whole install was under an hour on a login node, and it is the least interesting fact in this deck.**

Source: Instructor's shell log and harness state file, read 23 Sep 2026; times in the cluster's zone.

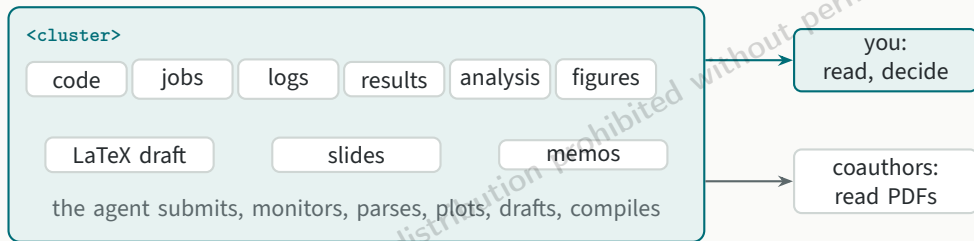
The Aha, on Tape: 5 May 2026



- ▶ “Before, I had it run the computation on the server, then I downloaded the results and analysed them myself.”
- ▶ “Then, almost by accident last week, I installed the AI on the server itself and let it operate there, adding its own checkpoints — and a lot of problems went away.”
- ▶ **The change was not a better model. It was where the loop ran.**

Source: The instructor, on a coauthor call, 5 May 2026; the instructor’s own words, translated. Nothing of the coauthor’s is shown.

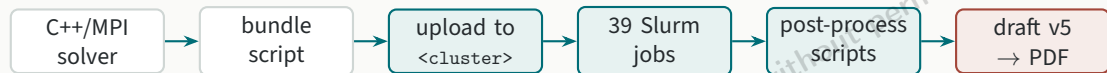
The New Loop: Everything Happens Where the Data Is



- ▶ One machine, one folder tree, one agent that runs from the project's root. Nothing is copied down to be analysed; what leaves the cluster is a PDF.
- ▶ **The loop that closed weekly closes the same day, and the researcher's part of it shrank to two verbs.**

Source: Instructor's project record, 2026, by role. A course-recorded run on the same cluster is left for the instructor to record; no number from it is printed here.

One Project, End to End: The National-Debt Pipeline



each arrow is a command the agent runs from the project root;
the draft's numbers trace to job outputs; the coauthor reads PDFs

- ▶ The model is a model of government debt and default, solved by sweeps of C++/MPI jobs; the 2026 revision programme was 39 jobs, designed from a coauthor's comments and submitted within days of them.
- ▶ The project's standing brief lists every script and what it does, so the agent can rerun the chain without being told how.
- ▶ **The unit of instruction became a project state (“follow the plan; run the new experiments there”), not a question.**

Source: Instructor's project repository, read 23 Sep 2026; paths symbolic; no result shown.

Four Real Prompts, Redacted

4 May 2026, a literature project [monitor]

> i just worried that <cluster> may kill the jobs if they run more than 30 minutes or so. keep an eye on that. keep doing what you are doing.

25 Jul 2026, the national-debt project [execute the plan]

> Then follow the plan, make changes. For new experiment, just keep in mind, I will run them on a different cluster, <cluster>. See here for previous script how we run experiments. <submit script>

25 Jul 2026, the national-debt project [diagnose]

> mkdir: cannot create directory '<spool path>': Permission denied.
all jobs failed due to errors above.

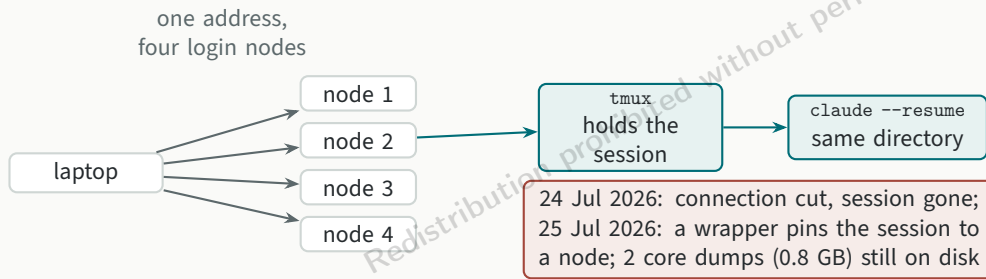
25 Jul 2026 [govern the tool]

> use job-specific scancel <jobid> instead of scancel -u <user>, so stops killing unrelated jobs.

- **The verbs are monitor, execute, diagnose, govern. None is a question; each hands over a state of the project and a rule.**

Source: Instructor's harness prompt history, 2026; typed as shown; names of hosts, scripts and paths redacted.

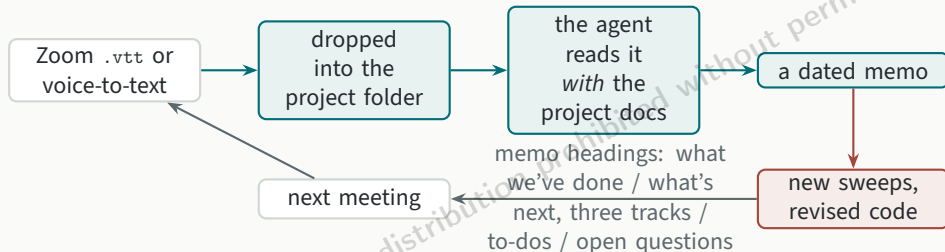
Things Break: Surviving a Disconnect



- ▶ “The problem is when the connection to cluster was cut off ... I cannot resume the previous session. Is there any way to avoid that?”
- ▶ **A session is a list on disk (Deck 6.2), so it resumes from the same directory — but only on a node you can find again.**

Source: Instructor’s notes file and wrapper script, 24–25 Jul 2026; harness history; counts by role, 23 Sep 2026.

The Meeting Loop



one instance, by file times: transcript 02:29 → memo 03:44 → document 03:46

- ▶ The memo is a file in the repository, so the next experiment is designed from a document, not from memory.
- ▶ **Seventy-five minutes from a recording to a memo the coauthor can read — and the memo names the next sweep.**

Source: Instructor's memos of 5 May and 7 Jun 2026 and one interview pipeline of 20 Jun 2026; headings and file times only.

From Results to a Draft in One Day

status memo

- 1 TL;DR
- 2 cleanup
- 3 toolchain
- 4 sweep inspection
- 5 rerun (job ids)
- 6 next steps

a 57-page
computational draft,
10 Sep 2026

its provenance rule:
“no number appears in
the paper that is not
traceable to a frozen artifact”

the memo's results table is shown with its cells blanked:
what matters is that a table with job ids and pre-flight checks exists before a sentence of the draft does

- ▶ The draft was written from a plan file in a day; every figure regenerates from frozen artifacts by a script.
- ▶ **Speed is not the point: a fast draft is only safe when every number has an address.**

Source: A structural-model project's records, by role: memo structure, page count, date and the rule; no result shown.

The Inventory: What the Agent Does for Me Now

coding	mathematical analysis	numerical design	empirical analysis
editing drafts	formatting slides and drafts	publishing the website	literature review
IT: job errors, package installs	meeting memos	continuous learning: the tool teaches its own failure modes	

- ▶ IT without a ticket: a submission error parsed and fixed, a CPU job that must still request a GPU, a TeX distribution installed into project space — each one prompt and one afternoon.
- ▶ **The mundane list is the biggest saving; the last item is the one that changes the researcher.**

Source: Instructor's own account, Aug 2026; a status memo of 9 Jun 2026, \$3, and a job script; artifacts by role.

Where the Time Went: One Week, Before and After

before

write and debug code

move files, wait

analyse

write

meet

after

specify

read diffs, logs, drafts

decide, verify

write

meet

scale on the account, recounted 23 Sep 2026: 324,394 lines of Python, 135 experiment folders, 6,229 checkpoint files across two computational projects

- ▶ The strips are the instructor's description, not a time-use survey; the counts are the recount, with their definitions.
- ▶ **What grew is not the number of experiments alone. It is the share of the week spent reading what a machine produced.**

Source: Instructor's Aug 2026 statement, recounted on 23 Sep 2026 by role.

Continuous Learning: The Log as a Textbook

a project log, 861 KB

139 entries, dated

...

2026-07-22 — probe verdict

2026-07-22 — probe 2

FAILED fast

2026-07-24 — gate FAILED,
cause identified

SUPERSEDED — read the
resume file first

every correction
becomes a file

the tool learns the
project's conventions

the researcher learns
the tool's fail-
ure modes

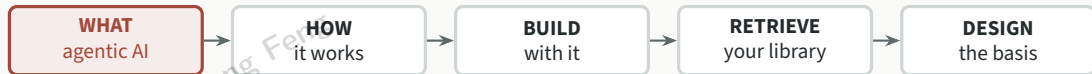
- ▶ Entries in capitals are verdicts: a probe that failed, a gate that failed, a plan that was superseded. They are described here, not quoted.
- ▶ **The log is where the two kinds of learning meet, and it is the artifact a new coauthor reads first.**

Source: A project log, by role; counts on 23 Sep 2026.

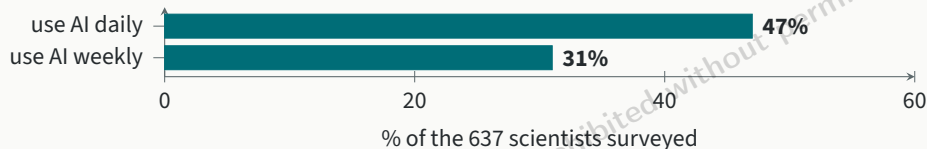
Session 6.1

Where the Value Shows

637 scientists said where their saved time went — Session 1's survey, re-read with one question



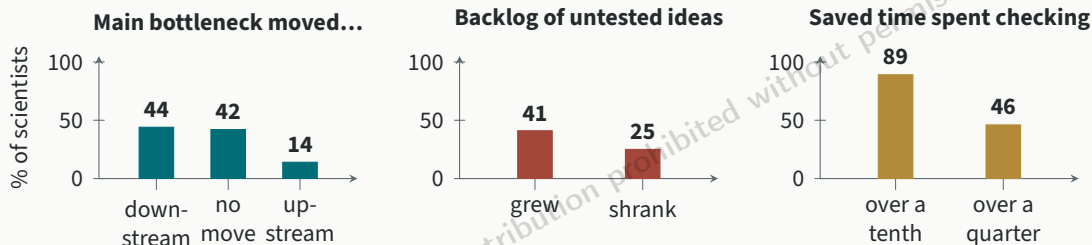
What a Sample of 637 Scientists Reported



- ▶ Nine days ago, in Deck 1.1, this survey showed that scientists use AI far more than other occupations. Today it is re-read with one question: *where did the saved time go?*
- ▶ The base: 637 active researchers in the US and UK, fieldwork 27 July to 11 August 2026; a screened non-probability sample.
- ▶ 47% use some form of AI daily and another 31% weekly. Adoption is no longer the question.
- ▶ **The next two frames are Session 1's, redrawn as they were; each carries one new bullet, and it is this deck's.**

Source: Codreanu, Imas, Mateos-Garcia et al. (2026), *AI in Science: Early Insights*, pp. 4, 23, 42; read 23 Sep 2026.

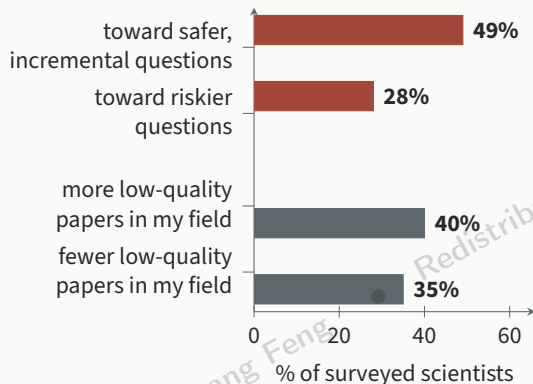
Findings 3–4: Time Saved, and Where the Bottleneck Went



- ▶ Reported saving: 6.9 hours a week; just under three quarters save time on net, 6% lose it. About 70% of it returns to research: more output (just under 30%), lab work (21%), harder problems (19%).
- ▶ **The verification tax: of those who save time, 89% spend over a tenth of the saving checking AI output, and 46% over a quarter.**
- ▶ **Part D's first danger, in survey numbers: the queue moved to the checker's desk.**

Source: Codreanu et al. (2026), pp. 5, 23–26 (their Figure 13). Third panel: those who report saving time.

A Warning Inside the Same Data



- ▶ 84% report more output over three years. Output of what kind?
- ▶ The authors' reading: perhaps a "Streetlight Effect". AI cheapens work where data and benchmarks already exist.
- ▶ Top bottleneck today: physical experiments and data collection (24%). No language model runs a laboratory.
- ▶ **More output is not the same as more discovery.**

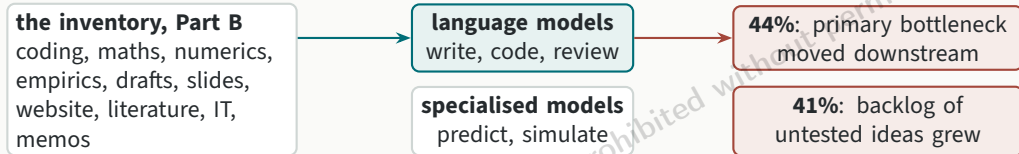
Source: Codreanu et al. (2026), pp. 24, 26, 30.

How to Read This Evidence

- ▶ The usage data are the vendor's own telemetry. Enterprise accounts and agentic tools are not observed.
- ▶ The survey is a screened non-probability sample: unweighted, with no margin of error quoted.
- ▶ Enthusiasts answer surveys about AI. The authors say this may overstate adoption and time saved.
- ▶ Every quantity is self-reported, and the results “establish associations rather than causality”.
- ▶ **What the data do support:** scientists use AI far more than other occupations, and they report a constraint moving from generating ideas to testing and checking them.
- ▶ **Self-reports against a measurement** (METR, Jul 2025): developers forecast 24% faster, believed 20%, measured 19% *slower*, 9% of time reviewing AI output; METR's Feb 2026 follow-up (−4%, −15 to +9) is, it says, an unreliable signal — selection.

Source: Codreanu et al. (2026), §6.1 (limitations), pp. 28–29, and appendix, p. 42.

The Cluster Story in Their Numbers



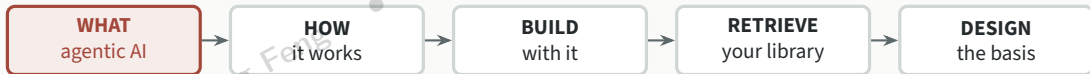
- ▶ The report's two families: language models that write, code and review; specialised models that predict and simulate. Every item on Part B's inventory is in the first; the solver is C++, not a model.
- ▶ 44% say their primary bottleneck moved downstream in two years; 41% say their backlog of untested hypotheses grew. Part B's meeting loop ended in "new sweeps, revised code".
- ▶ **"Suggest new sweeps" is the backlog: proposals arrive faster than one researcher can judge them. One account, one draw from this population — and not an outlier.**

Source: Codreanu et al. (2026), pp. 4–5, 25; Part B's inventory and meeting-loop frames.

Session 6.1

The Researcher as Project Manager

What the job became, and two dangers named now



The Bottleneck Moves; Four Judgements Do Not

is the *question*
worth asking?

is the *model*
the right one?

is the *design*
credible?

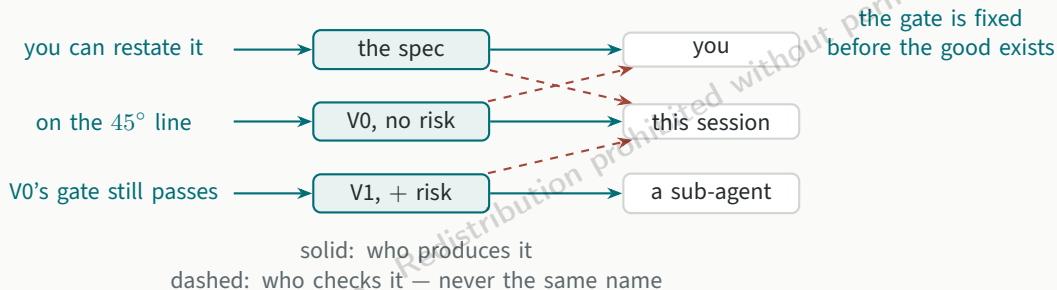
is the *result*
believable?

None of the four is cheaper than it was. Each is now a larger share of what you are paid for.

- ▶ Deck 1.1 said it in prices: when one input becomes elastic, its complements bind. The four judgements are the complements.
- ▶ Execution was the bottleneck: the weeks between a good idea and a first number. When that cost falls, the scarce input is knowing which numbers to believe.
- ▶ **In the year on the cluster, the researcher's two verbs became “read” and “decide” — and both got longer, not shorter.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.1; Deck 1.1's frame on elastic inputs.

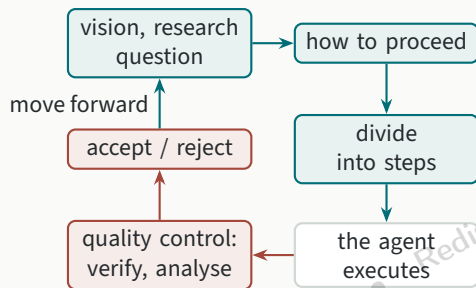
You Are the Project Manager



- Break the goal into **intermediate goods** — a specification, a validated V0, a cleaned panel, a review report — and fix each good's gate *before* it is produced.
- **Delegating is a principal-agent problem, so every good gets a gate, an owner, and a reviewer who did not build it.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.1 and §4.4, the agent workflow.

The Job Description



A **gate** is a bound fixed before the work exists: a number to reproduce, a test to pass, a property to hold. Written after the fact it is not a gate but a rationalisation.

- ▶ Six of the seven boxes were always the researcher's. What changed is that the seventh no longer takes the weeks.
- ▶ **The job is the cycle, and the cycle is management.**

Source: Instructor's account, 2026; Miao & Feng, *AI for Economists*, ch. 15, §4.1.

A Paradigm Shift: What Changed, What Did Not

What changed (the national-debt project's record)

- ▶ The comment–revise–rerun loop compressed: a coauthor's comments of 22 Jul, a response plan by 24 Jul, 39 jobs submitted from 25 Jul.
- ▶ Documentation became executable: the standing brief lists every script and what it does, and the agent runs from it.
- ▶ The unit of progress became the *validated milestone*: a sweep that passed its checks, not a week that passed.

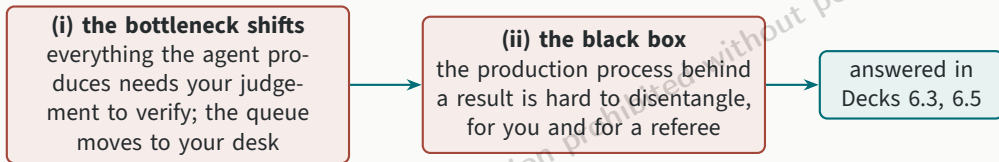
What did not change

- ▶ The AI never decided what was economically interesting.
- ▶ It never caught the wrong timing on its own.
- ▶ It never knew when a result was not credible.

The machine moved every bound except the one that defines the researcher.

Source: Instructor's project record, file dates read 23 Sep 2026; the right column from the instructor's earlier lecture notes.

Two Dangers, Named Now and Answered at the End

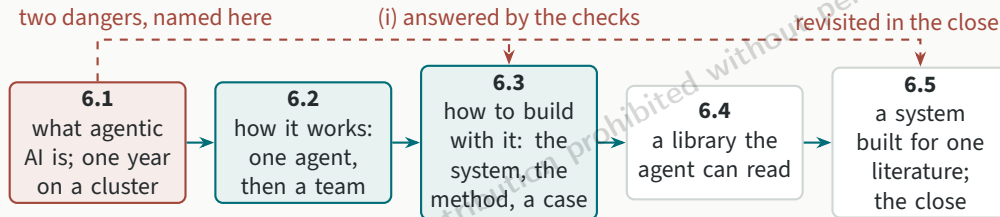


the survey evidence for (i) is Part C: 637 scientists, and where their saved time went

- ▶ Both are real on the account described here; neither is an argument for going back.
- ▶ **For now the priority is to understand how to leverage the rapid advance so that it works for us — and to name what it costs.**

Source: Instructor's account, Sept 2026; Codreanu et al. (2026) as the pointer to the evidence.

The Session Map: Five Decks, One Arc



- ▶ Decks 6.1 to 6.3 follow one thread: what makes AI agentic, how it works, how to build with it. Then a library the agent can read (6.4), and a system built for one literature, with the close (6.5).
- ▶ **Deck 6.2 next: a loop you could write yourself, and the four marks of an agent as lines of code.**

Source: The session's five decks; Miao & Feng, *AI for Economists*, ch. 15.

Session 6.2

How Agentic AI Works: A Loop Built From Scratch, the Harness, the Session, and the Team

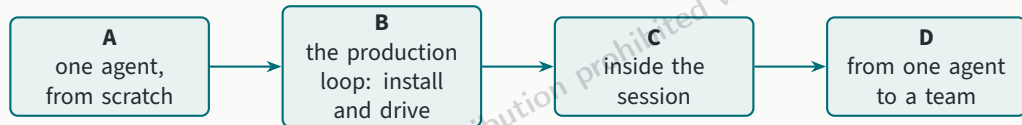
Part two of five: the loop in thirty lines you could write yourself, each of the four marks in code, the production harness that runs the same loop, what a session carries and costs, and how one agent becomes a team



Zhigang Feng

The Question, and the Route

How does agentic AI work — and could you write one yourself?



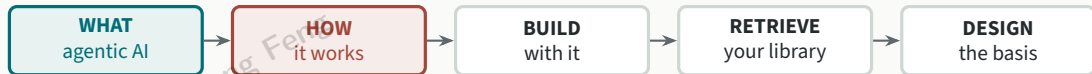
- ▶ A working definition: “An LLM agent runs tools in a loop to achieve a goal.”
- ▶ Deck 6.1 named four marks — tool use, autonomy, iteration, persistence. Part A puts each one into code; the running task is official uninsurance rates by age.
- ▶ **An agent is a model, a loop and tools; everything else is rules and files.**

Source: S. Willison, “I think ‘agent’ may finally have a widely enough agreed upon definition to be useful jargon now”, 18 Sep 2025; Miao & Feng, *AI for Economists*, ch. 15, §3.3.

Session 6.2

One Agent, From Scratch

One call, one loop, one tool, three rules, memory and a verifier: the machine in thirty lines



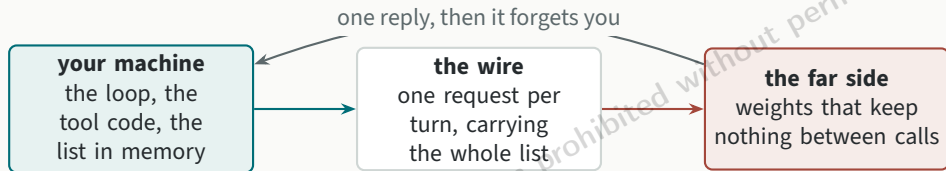
Step 0: The Simplest Model Call, and What It Cannot Do

```
messages = [  
    {"role": "user",  
     "content": "Plot uninsurance  
                by age, 2008–2024."}  
]  
reply = model(messages)  
print(reply.text)
```

- ▶ What comes back is *a description of what you should do*, possibly with code in it.
- ▶ No file was read. No command ran. Nothing on disk changed.
- ▶ **The model has no hands. Everything in this deck is about what happens when you give it some.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.1.

Step 1: The Loop, Not a Place



- ▶ Plan, act, observe, repeat: the model decides; the harness executes and reports back; the model reads the report and decides again.
- ▶ An error message is not a failure of the loop; it is the observation the next step is planned from.
- ▶ Two of Deck 6.1's four marks live here: autonomy and iteration.
- ▶ **The loop stops when the model stops asking for tools — not when the task is correct.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.3.

Step 2: A Tool Is a Typed Function

```
{ "name": "score_forecast",  
  "description":  
    "Score a forecast of the  
    ten-year yield on the  
    frozen test window.",  
  "input_schema": {  
    "type": "object",  
    "properties": {  
      "path": { "type": "string",  
        "metric": { "enum": ["rmse",  
                           "mae"] } },  
      "required": ["path"] } }
```

- ▶ A tool is a function the model may call: a name, a schema for its arguments, a description.
- ▶ The model does not run your code: it writes one line of JSON; the harness runs the function.
- ▶ Ill-typed arguments are refused before they run; types are checked, meaning is not.
- ▶ **The description is the only part the model reads when deciding to call it at all.**

Source: Model Context Protocol specification, revision 2026-07-28, tools; Miao & Feng, ch. 15, §4.3.

Step 3: The Whole Loop, in a Dozen Lines

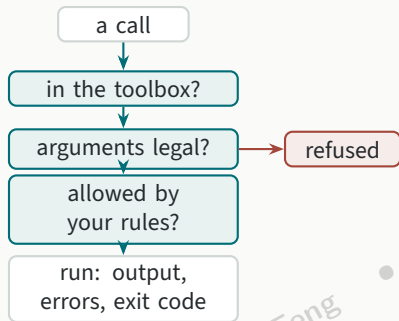
```
messages = [user("Plot uninsurance by age.")]

while True:
    r = model(system=RULES,      # standing rules
              tools=TOOLS,      # names only
              messages=messages) # the whole list
    if r.stop_reason != "tool_use":
        break                  # nothing to run
    messages.append(assistant(r.content))
    out = [run_tool(c) for c in r.content
           if c.is_tool_use]
    messages.append(user(out))
```

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.3; Yao et al. (2023).

- ▶ A person writes this file, once; the model never writes or runs it. The literature calls the loop **ReAct**: reason, act, observe.
- ▶ What the model chooses: which tool, which arguments, and when to stop asking.
- ▶ **The list only grows, and is re-sent whole on every pass. That line is your cost curve.**

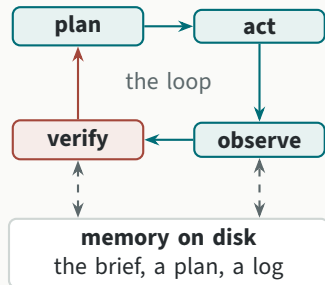
Step 4: Three Rules Around the Loop



- ▶ Fail any one and the call is refused — and the refusal goes back to the model as an observation, which is how it learns the boundary.
- ▶ The lab's loop writes the rules as code: an allow-list, a gate that can block a write, and a budget on turns and tokens.
- ▶ **A syntax line at the head of an ado-file does this job already: an illegal call is stopped before it computes.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.3; the session's lab, block A.

Step 5: Memory and a Verifier — Plan, Act, Observe, Verify



- ▶ **Orchestrator**: the loop from Step 3; the model makes every decision in it.
- ▶ **Memory**: files read at the start and written as it goes; the lab's loop writes a table only with its source.
- ▶ **Verifier**: a test, a benchmark or a reviewer that decides what survives.
- ▶ **Persistence and checking complete Deck 6.1's four marks: each is now code or a file.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.3; the session's lab, block A.

Run It: Seven Turns, Traced

turn	what it called	what came back
1	rank the statements: inflation, March 2022	three candidate meetings
2	that meeting's sentences on inflation	the quotation, and the statement's URL
3	the yield over the following month	2.32 to 2.89 per cent
4	a shell command, to read the panel itself	refused: not on the allow-list
5	write the answer as a table	blocked: the table names no source
6	the same write, now carrying its source	written
7	the answer: quotation, URL, window, change	end of turn

- **The rules act on two of the 7 turns. The trace is the deliverable; the answer is what it happens to contain.**

Source: The session's lab, block A, the replayed run.

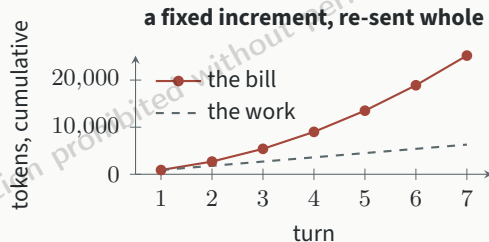
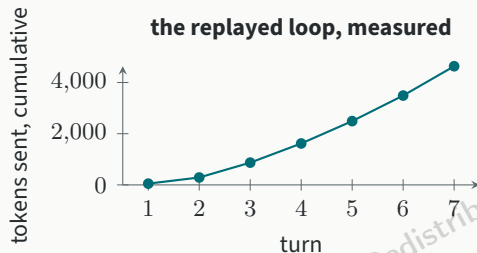
The Session Is a List, Not a Connection



- ▶ The endpoint keeps nothing, so each turn is a fresh request carrying the whole conversation so far.
- ▶ “The window is full” means this list outgrew the limit — and it grows fastest from tool output, not from what you type.
- ▶ **Clearing a session and compacting it are both operations on a list.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.1.

The Session Is a List, So the Bill Grows Faster Than the Work

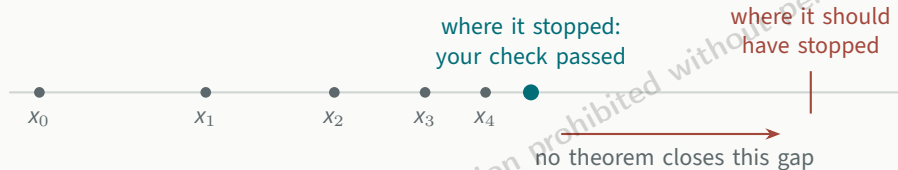


- ▶ A turn appends to the list and re-sends all of it: the work grows by the increment, the bill by the length. Under a budget it can stop mid-task, with no answer.
- ▶ **Sum a list that grows linearly and you get a square: twice the turns, four times the tokens. Files and short sessions bend it back.**

the lab's replayed loop, a fixed script in place of the model, tokens counted at four characters each · built 20 September 2026 ·
the lab's bounded task; exported 23 September 2026

Source: Course-recorded replay, exported by the lab's exporter; Miao & Feng, *AI for Economists*, ch. 15, §3.3.

An Iteration Without a Convergence Theorem



- ▶ The loop is an iteration in the ordinary numerical sense: no closed form, just the newest value used immediately by the next step.
- ▶ Stopping is declared by something testable — a test passes, an exit code is zero, an output matches a benchmark fixed beforehand.
- ▶ **Gauss–Seidel has conditions under which it converges; this loop has none. It can settle, with confidence, on a wrong answer that passes a wrong check.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.3, on the loop and its disanalogy.

From Thirty Lines to a Production Harness

in your thirty lines

the message list
the system text
the tools
the allow-list, the gate
the budget
a second loop
a prompt loaded on demand
propose, do not act

in Claude Code (and every harness like it)

the session: resume it, clear it, compact it
the standing brief (CLAUDE.md) and rules, loaded at every start
Read, Edit, Bash, Grep — and servers on one open protocol
permission rules, and hooks that can refuse a call
the meter: context used, tokens spent
a sub-agent, with a list of its own
a skill
plan mode

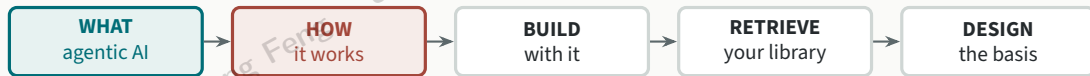
- ▶ The model decides what to run; the harness runs it; R or Stata still does the estimation.
- ▶ **The weights never change. What the harness adds is closed feedback, persistence, and observation over recall.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.2–§3.3, and its thinkbox on what a harness changed; harness command-line help, course machine, 2026-09-20.

Session 6.2

The Production Loop: Install and Drive

The same loop on a laptop and on a login node, and a first hour that ends in a commit



Install Once: Three Durable Steps

```
# 1. install the CLI, once
#   per machine
$ curl -fsSL https://claude.ai/\
    install.sh | bash

# 2. authenticate, once
$ claude
#   a browser opens; sign in

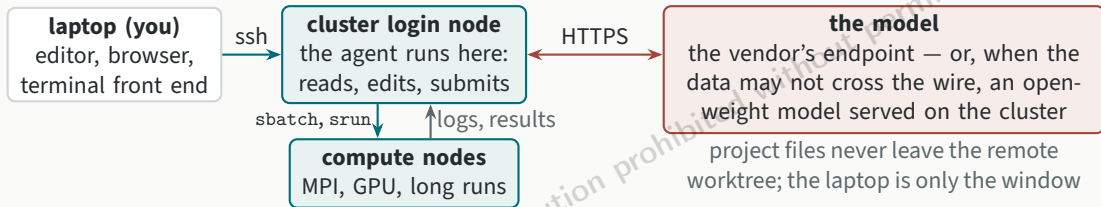
# 3. work from the project,
#   every session
$ cd my-project
$ claude
```

Install once per *machine*. Authenticate once per *account*. Start from the project folder every session — that folder is the agent's whole world.

- ▶ Only step 1 differs between tools: Codex CLI, Antigravity CLI and Qwen Code run the same loop and speak the same tool protocol.
- ▶ **Versions on the course machine on 20 September 2026: 2.1.278, 0.155.1, 0.47.0.**

Source: Vendor installation documentation; versions probed on the course machine, 20 September 2026.

Move the Agent to the Compute: The Cluster Install



- ▶ No `sudo`: node via `nvm` or `module`, the binary in `$HOME/.local/bin`, and sign-in prints a URL you open on the laptop (`claude auth login`, read from its help and not run).
- ▶ **The rule: install where the data, modules, GPUs and job logs live.** A local agent copies cluster files back and forth; fine for toy code, fragile for research.
- ▶ Orchestrate on the login node, compute through the scheduler: centres stop agent processes there “without prior notice”. Work inside `tmux`.

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.2; Anthropic, “Long-running Claude for scientific computing” (23 Mar 2026); Aalto, “AI Agents on HPC”, 23 Sep 2026.

Git Is the Safety Net



- ▶ An agent that can write files can also overwrite them. Version control turns that from a disaster into an inconvenience.
- ▶ Commit before you hand over the keyboard, and again whenever a step worked. `git diff` is the audit: read it before you believe a report.
- ▶ **Without it you are asking a stranger to edit your only copy.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.2.

Your First Session: Read First, Plan Before Code

```
$ cd fomc-yields  
$ git add -A && git commit -m wip  
$ claude
```

```
> Read the files here and tell me  
   what this project does. Do not  
   change anything yet.
```

- ▶ The first instruction is read-only on purpose. You are checking whether it understands the project, not whether it can type.
- ▶ Then **plan mode**: it reads and proposes, and writes nothing until you approve. The plan is a file; resume from it, not from the transcript.
- ▶ **Ask it to restate the task in its own words. Disagreement is cheapest here.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.2.

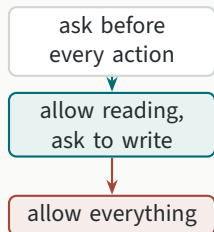
Commands That Earn Their Keep

Command	What it does, and why it earns a place
/init	writes a first standing-instructions file from the repository — the brief re-read at every start
/plan	enables plan mode: it explores and drafts, and changes nothing until you approve
/model	switches model mid-session, context kept: a derivation and a rename do not deserve the same engine
/effort	sets the effort level for the session — the largest quality-against-cost dial you hold
/context	draws context use as a grid: the invisible constraint, made visible
/compact	summarises the session so far — one you choose beats one that fires mid-derivation
/rewind	restores the conversation, or the files, to a checkpoint (Escape twice)

Four dials: what it reads, how hard it thinks, how much it may do unasked, what it costs.

Source: Harness command-line help and installed packages, course machine, 20 September 2026.

Permissions: The Blast-Radius Dial



only in a
container you
can throw away

Where to set it is a rule, not a mood: Deck 6.3 names it — least privilege — and says which three things an agent must never hold at once.

- ▶ Irreversible actions — deleting, sending, spending — are confirmed one at a time.
- ▶ **The dial is the main safety control you have, and it is yours to set per task.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.4.

Your First Hour: A Walkthrough Checklist

step	the command	what you look at
1 install	<code>curl -fsSL https://claude.ai/install.sh bash</code>	claude --version answers
2 version control	<code>git init && git add -A && git commit -m "before"</code>	a clean tree to fall back to
3 the brief	claude, then <code>/init</code> ; edit the file it wrote	goal, rules, one gate
4 plan	<code>/plan</code> : “restate the task; propose steps; change nothing”	does it understand the project?
5 first task	one read-only task, then one small write	the transcript
6 read the diff	<code>git diff</code>	every changed line, before any report
7 commit	<code>git add -A && git commit -m "first agent task"</code>	a checkpoint with your name on it
8 the meter	<code>/cost</code> (and <code>/usage</code> for the plan’s meter)	what the hour cost, before the day does

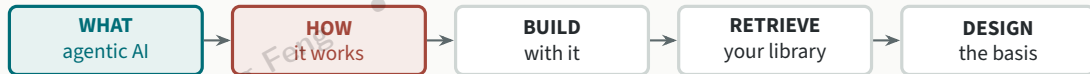
Eight steps, and only 3, 4 and 5 involve the model. The rest is the discipline you already had, done before and after. Steps 2, 6 and 7 were executed in a scratch folder on the build date; the harness commands were read from its help; no agent was launched for this deck.

Source: Course machine, 23 Sep 2026; Miao & Feng, *AI for Economists*, ch. 15, §4.2.

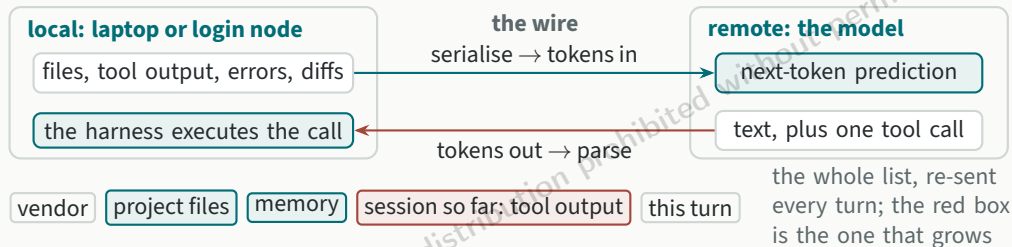
Session 6.2

Inside the Session

What crosses the wire, what fills the window, and what compaction forgets



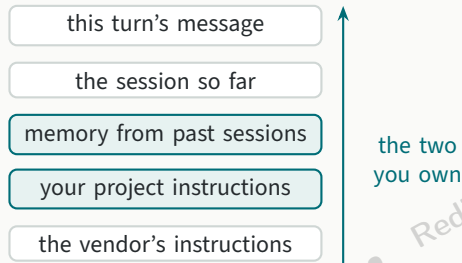
What Crosses the Wire, Drawn With Tokens



- ▶ Nothing of yours moves. The model never logs in; what crosses is text — a tool call out, an observation back — as tokens.
- ▶ Any file it reads crosses too: what it may open is an authorisation decision.
- ▶ **The meter counts both directions, and the part of the list that grows is what your tools printed.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.2–§3.3.

Five Layers of the Context Window

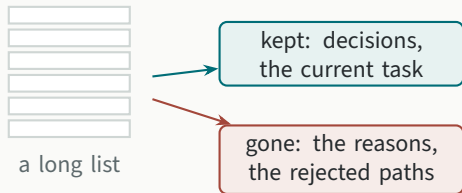


loaded bottom to top,
before you type

- ▶ Every turn, the model decides from one place only: whatever is in this window now.
- ▶ The first layer and the last are not yours to manage. **The middle two are, and both are files on disk.**
- ▶ The window is finite and a long session will exhaust it — but the binding constraint is attention, not size.

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.3.

Compaction: What It Keeps, and What It Drops



standing instructions and rules
are re-read from disk

- ▶ When the window fills, the harness summarises the session and carries on. It happens on its own unless you do it first.
- ▶ Lossy in a specific way: the conclusions survive; the reasons behind them, and the small corrections you made in passing, often do not.

State belongs in files, not in the conversation. Write down anything that must outlive the window, and compaction costs you nothing.

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.3; harness help, course machine, 20 September 2026.

Context Rot, and Why Starting Costs Something

loaded grows every turn

the fixed cost is paid before you type



stale observations
crowd out what matters

a correction made three
hours ago can vanish

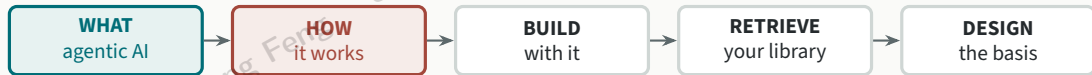
- ▶ **Context rot:** the longer a session runs, the more failed attempts and stale output sit between the model and the thing that matters.
- ▶ The remedy is management, not patience: summarise a finished stage, and start clean for an unrelated task.
- ▶ **Starting is not free either — instructions, rules and memory load first. Short files, short sessions.**

Source: Liu et al. (2024), *TACL* 12; Miao & Feng, *AI for Economists*, ch. 15, §3.3.

Session 6.2

From One Agent to a Team

Where the reasoning lives, how a second loop starts, and the patterns teams are built from



Tool, Skill, Sub-agent: Sorted by Where the Reasoning Lives

tool

code you wrote
the model picks
the *arguments*
no reasoning inside

costs a schema
fails by raising an error

skill

prose you wrote
the model picks *to read it*
reasoning stays
in the main loop

costs one description line
fails by being ignored

sub-agent

a second loop
its own context and tools
reasoning happens
elsewhere

costs a second context
fails by drifting off-task

- ▶ Loading order is cost order: rules load at every start, a skill's body only when its description matches, a sub-agent in a fresh list.
- ▶ You will read that a skill is “a deterministic piece of software with no brain of its own, which cannot decide when it should be used”. An accurate description — of a tool.
- ▶ **Ask of any new agent term: once it is installed, where does the reasoning happen?**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.3.

A Skill Is a Folder the Model Opens Only When Needed

always in the window
the description line



on trigger
the one-page manual is read



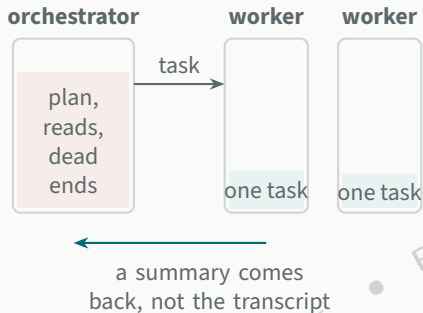
on demand
references load, scripts run

cheap to keep twenty; you
pay for the one you use

- ▶ Three parts: a one-page manual whose description says *when* to call it; scripts for what must be deterministic; references with worked examples.
- ▶ The folder format is an open standard (agentskills.io, published December 2025, originally developed by Anthropic), read by a growing list of agent products — so a lab's skills outlive a vendor.
- ▶ Deck 6.1's meeting loop is one; Deck 6.4 wires another to a real index.
- ▶ **The container is for a team's working knowledge, not for a task.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.3; Agent Skills specification and Anthropic's announcement of 18 Dec 2025, read 23 Sep 2026.

Isolated Context, and Dispatch by Description



- ▶ A sub-agent is not extra intelligence — the model is the same. It is a second, empty list.
- ▶ The worker never sees your dead ends, so it cannot be misled by them, and your list grows by a summary rather than by everything it read.
- ▶ Dispatch is a tool call chosen from your descriptions.
- ▶ **A clean window buys focus, not independence: the worker still inherits your framing.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.3; Anthropic engineering, multi-agent research systems (13 June 2025).

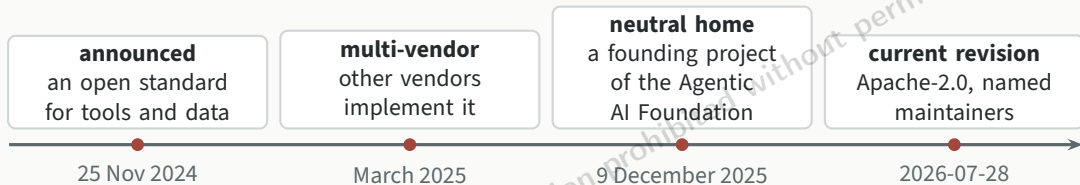
Four Orchestration Patterns, and Which One Claude Code Runs

pattern	shape	in Claude Code
orchestrator-workers	a lead splits the task and dispatches workers	the session dispatching sub-agents
reflection	an output is critiqued and revised before it returns	a reviewer agent; Step 5's verifier
plan, then execute handoff	one pass plans, a second carries it out control passes to a specialised agent	plan mode, then the build a plan file the next session picks up

- ▶ Handoffs “are represented as tools to the LLM”: dispatch is a tool call chosen from your descriptions, and editing one stops it.
- ▶ **The same loop, wired differently. Which wiring your project needs is Deck 6.3’s question.**

Source: Pattern names: Anthropic, “Building effective agents”, 19 Dec 2024 (orchestrator-workers); A. Ng, *The Batch*, 27 Mar 2024; OpenAI Agents SDK docs, handoffs; Claude Code docs, permission modes; all read 24 Sep 2026.

One Protocol for What Lies Past Your Disk



- ▶ Deck 1.3 put “one protocol” on the 2024 mark. This is it: the Model Context Protocol.
- ▶ The neutral home is the Linux Foundation’s **Agentic AI Foundation**, founded with the protocol (from Anthropic), goose (from Block) and AGENTS.md (from OpenAI).
- ▶ So MCP is one of three standards you already use, with the brief file and the skill folder (f25); pin its revision as you pin a package.
- ▶ **Cite the revision, not the vendor: as of 20 September 2026 it is 2026-07-28.**

Source: Anthropic, “Introducing the Model Context Protocol” (25 Nov 2024); Linux Foundation press release, 9 Dec 2025; the protocol’s governance and versioning pages, read 20 September 2026.

What You Now Understand, and What Deck 6.3 Builds With It

tool use
a typed
function

autonomy
the loop
decides

iteration
observations,
refusals

persistence
files on disk

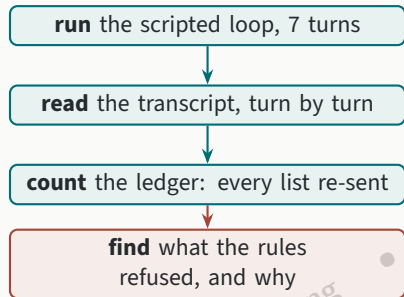
a team
sub-agents,
patterns

checks around all of it: an allow-list, a gate, a budget, a verifier

- ▶ You wrote the loop, ran it offline, and met the same loop in Claude Code, on a laptop and on a login node.
- ▶ You know what one session carries and costs, and how one agent becomes a team.
- ▶ **Deck 6.3 builds with it: a repository that starts itself, a team you can check, and a method for growing a project.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §3.3 and §4.3.

Your Turn: Lab Block A, the Loop Offline



- ▶ Self-paced, after class, offline: a CPU, no account, a clean kernel, fixed seeds. About twenty minutes.
- ▶ The loop's model is a script, not a network call, so your transcript and the recorded one agree line for line.
- ▶ **Block A is Part A of this deck, run: the same classes, the same rules, the same bill.**

Source: The session's lab, block A, and its recorded-results files.

Session 6.3

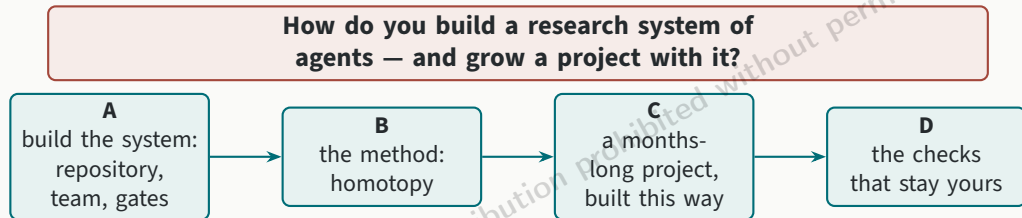
How to Build With Agentic AI: The System, the Homotopy Method, a Research Case, and the Checks

Part three of five: set up a research system of agents you can trust — the repository, the team, the gates — then grow a project with it one validated margin at a time, as a months-long research project did



Zhigang Feng

The Question, and the Route



- ▶ Deck 6.2 showed how the machine works. This deck is what you build around it, and how you work with it.
- ▶ Two scales of one discipline, both from the instructor's work: a twenty-two-minute task and a months-long research project.
- ▶ **The system is set up once; the method is used every day; the checks never go away.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.1–§4.3.

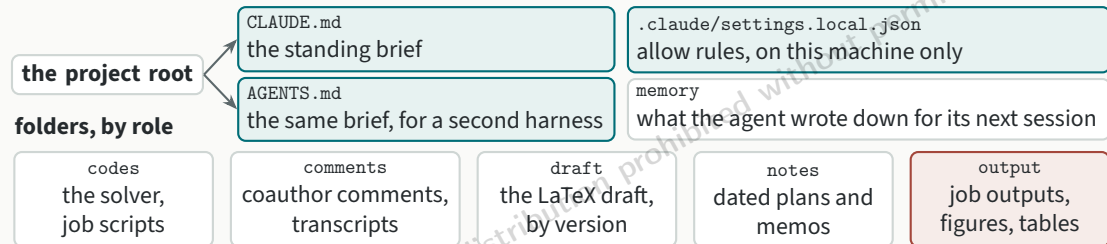
Session 6.3

Build the System

The repository, the team and the gates: what you write before the first prompt



The Repository Is the Agent's Memory



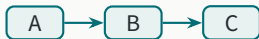
every arrow of Deck 6.1's pipeline starts in one of these folders and ends in another; the two files at the top are what let a session that knows nothing pick the chain up

- ▶ The national-debt project's root, by role. Your home folder holds your defaults; the repository holds what a coauthor clones, and where they disagree the project wins.
- ▶ **The harness reads these files first at every start, so a session that knows nothing picks the work up from them.**

Source: Instructor's project repository, read 23 Sep 2026; folder names by role; no content shown.

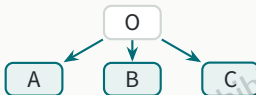
Design the Team: Three Wirings, and Who Checks Whom

sequential



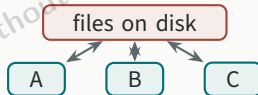
each list seeded
by the last output

hierarchical



workers report up;
only O sees all

blackboard



all read and
write one store

- ▶ Strip the vocabulary away and multi-agent designs differ in one respect: who writes into whose list.
- ▶ Two names for every good: who makes it, who checks it. The producer is never the only reviewer; a review panel is the parallel wiring, its reports kept apart.
- ▶ **In a research project the blackboard is files on disk — which is why a file-driven workflow is the one that survives a restart.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.3–§4.4.

One Agent or a Team? Marginal Value Against Marginal Tokens



- ▶ One vendor's own eval: a lead agent with workers beat a single agent by 90.2%, and multi-agent runs use about $15\times$ the tokens of a chat.
- ▶ The quota is tokens per minute, not agents; context binds per worker; and coordination — reading and reconciling n outputs — usually binds first.
- ▶ **Reach for more workers when the task is genuinely wide, not to feel parallel.**

Source: Anthropic engineering, multi-agent research systems (13 June 2025); Miao & Feng, *AI for Economists*, ch. 15, §4.3.

A Hook That Never Fires, and the Rule That Works

never fires

```
"PreToolUse": [  
  { "matcher": "Write(data/*.csv)",  
    "hooks": [{ "type": "command",  
                  "command": "check_schema.sh" }]}]
```

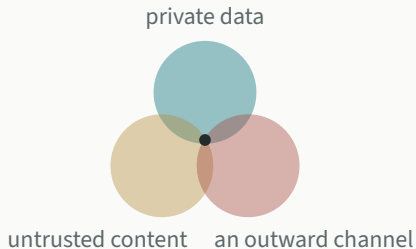
works

```
"PreToolUse": [  
  { "matcher": "Write|Edit",  
    "if": "Write(data/*.csv)",  
    "hooks": [{ "type": "command",  
                  "command": "check_schema.sh" }]}]
```

- ▶ A **hook** is a command the harness runs at a named event; before a tool call it can refuse the call, so it is a gate. The left one fails silently: a matcher with punctuation is read as a regular expression against the *tool name*, and no tool is named `Writedata`.
- ▶ Two fields, two jobs: the matcher selects on the tool's *name*; the `if` field filters on its *arguments*, in permission-rule syntax, and is only evaluated on tool events.
- ▶ **Test a gate by trying to break the rule it enforces. A rule you have never seen fire is a rule you have not got.**

Source: Claude Code hooks reference, matcher and condition fields, read 23 Sep 2026; the failure was observed on the course machine on 12 September 2026.

Least Privilege, Three Named Risks, and the Lethal Trifecta



“If your agent combines these three features, an attacker can **easily trick it** into accessing your private data and sending it to that attacker.”

- ▶ **Least privilege**: no write for a read-only task, no network for an offline one; match the grant to the task, not the model.
- ▶ The risk list: *prompt injection* first, *sensitive information disclosure* next; *excessive agency* climbed it in 2026.
- ▶ Anything the agent reads (e-mails, web pages, retrieved documents, even tool descriptions) is an untrusted injection channel.
- ▶ Data can reach the model’s instructions: with sensitive data, cut the outward channel.

Source: OWASP Gen AI Security Project, Top 10 for LLM applications (2025; 2026 revision) and Top 10 for Agentic Applications (Dec 2025); S. Willison, “The lethal trifecta for AI agents”, 16 June 2025.

Initialise a Research Repository in Eight Steps

step	the command, or the file	what it buys
1 a folder under version control	<code>mkdir savings-model && cd savings-model && git init</code>	one command back
2 the standing brief	<code>CLAUDE.md</code> : goal, three rules, one gate	the spec, re-read every start
3 least privilege	<code>.claude/settings.local.json</code> : an allow list of read and git commands	the blast radius, written down
4 one skill	<code>.claude/skills/weighted-rate/SKILL.md</code>	the weights rule, as an asset
5 one reviewer	<code>.claude/agents/reviewer.md</code> : reads, reports, never edits	a checker who is not the producer
6 one gate	a <code>PreToolUse</code> hook with the <code>if</code> field on data writes	a rule that refuses, not reports
7 the checkpoint	<code>RESUME.md</code> : state, and the next step	the next session starts here
8 the first commit	<code>git add -A && git commit -m "initialise"</code>	the tree a coauthor clones

Six files and two commands: a collaborator who clones it gets your agent, not your habits. Each skill or agent file needs a description that says when to use it.

Source: Course machine, 23 Sep 2026; Miao & Feng, *AI for Economists*, ch. 15, §3.3; Claude Code hooks reference, read 23 Sep 2026.

Session 6.3

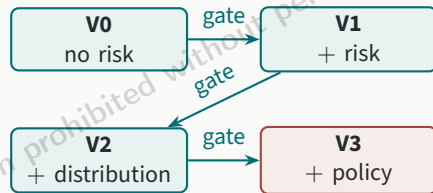
The Method: Homotopy

Solve the easy version first, validate it, then deform it toward the research task



The Homotopy Idea: Solve the Easy Version First

- ▶ Start from a version with clear intuition and a benchmark you understand.
- ▶ Use that validated version as the starting point for the next one.
- ▶ Add one margin at a time, and attribute a new bug to the margin you just added.
- ▶ **Never ask the agent to solve the full problem in one jump.**

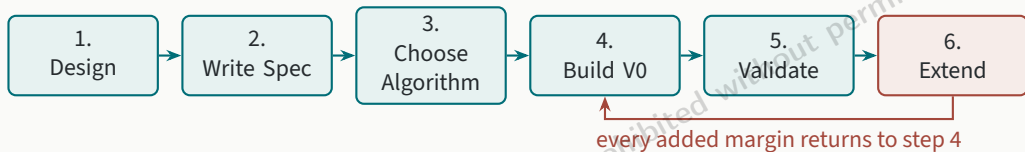


each step small, each step validated;
a failed step returns to the last version
that passed

Why the name: homotopy continuation starts from a problem with a known solution and deforms it continuously toward the target. V0 to V3 are gates along that deformation.

Source: Judd, *Numerical Methods in Economics* (1998), ch. 5; Miao & Feng, *AI for Economists*, ch. 15, §4.1.

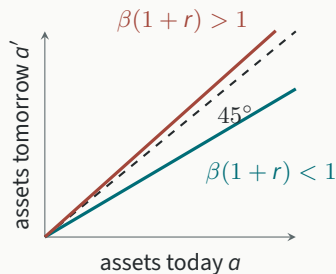
Six Steps, and the Spec Is the Contract



- ▶ A spec states four things: the assumptions and domain rules; the data or source restrictions; the procedure and its validation target; the outputs and where they are saved.
- ▶ Steps 1 to 3 produce no code. Ask the agent to restate the spec in its own words before it writes any.
- ▶ A specific request names its check: not “add some validation” but “an Euler-residual check, max and mean below 10^{-6} on the grid”.
- ▶ **The spec is the contract: the agent can be held to what it restated.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.1–§4.2; Judd (1998).

Step 4: V0, and the Gate Written Before the Run; Steps 5 and 6



- ▶ **V0**: the simplest version with a benchmark you fully understand — at $\beta(1+r) = 1$ the policy is *on* the 45° line. The three cases and a tolerance enter the spec first; nothing is added until V0 passes.
- ▶ Steps 5 and 6 climb $V0 \rightarrow V1$ (+risk) $\rightarrow V2$ (+distribution) $\rightarrow V3$ (+prices); a broken rung sends you back to the last one that passed.
- ▶ A machine checks convergence, NaNs, determinism; you check limiting cases, signs, magnitudes, economic sense.
- ▶ **Session 4's lab ran it once: seven blocks, ten gates fixed in advance, none a loss value.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.1 and §4.4; Session 4's lab notebook and its recorded gate file.

Grade the Plan Before Any Code: Five Criteria

1 it restates your goal, in its own words

2 steps you can review in minutes

3 a test per step, named in advance

4 stop and rollback points

5 the smallest testable object first

A plan without acceptance criteria is a wish. Send it back: a revision costs one message; a wrong build costs a day.

- ▶ Deck 6.2's plan mode shows you the plan; these five decide whether you approve it.
- ▶ In Part C, the first prompt applies the third criterion to the seed itself.
- ▶ **Approve against the criteria, not against the prose.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.2.

Validation Is Your Referee Checklist

workflow checks

convergence achieved
no NaN or Inf values
weights, mappings and source logs saved
residuals or quote checks pass
the script reproduces its output twice

economic checks

limiting cases recover known results
comparative statics make sense
magnitudes are plausible
inference uses the right errors and sample
you can say it in plain economics

- ▶ A machine can run the left column. Only you can run the right one.
- ▶ **If you cannot defend the output in a seminar, you are not done.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.4.

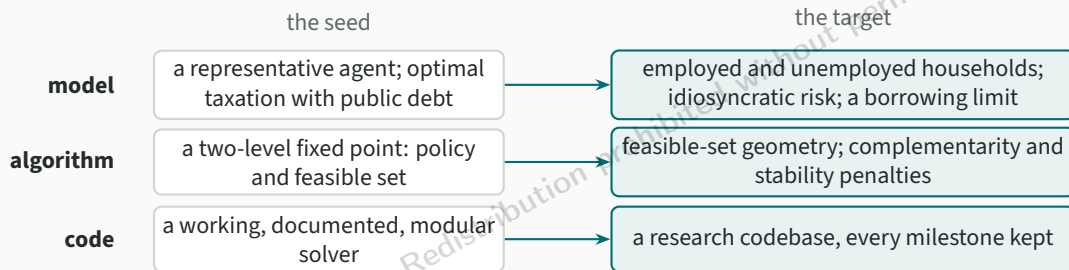
Session 6.3

A Months-Long Research Project, Built This Way

A validated seed, a team of sessions, a milestone ladder, three discoveries, and what stayed human



The Project: Where It Started, Where It Ended



- ▶ The seed was a working solver from an earlier paper; the target, an economy in which households need the debt to self-insure.
- ▶ **The agent compressed the loop between the three layers; it never chose the destination.**

Source: The instructor's research record (unpublished, coauthored), described by role; Miao & Feng, *AI for Economists*, ch. 15, §4.1.

What Makes a Good Seed

it works

reproduces its own paper, not just runs

it is documented

a TeX note in the code's notation

it is modular

each run a config diff, not a fork

it points ahead

the no-risk case of the target

- ▶ The classroom V0 and the research seed play one role: the thing every later failure is measured against.
- ▶ Model, algorithm and code deform together. Keep the TeX, the pseudocode and the code describing one economy: drift between them is where silent errors live.
- ▶ **A good seed is to agentic research what a good instrument is to empirical work.**

Source: The instructor's research record, described by role; Miao & Feng, *AI for Economists*, ch. 15, §4.1.

Opening the Conversation: Restate, Then Audit

1. "Here are the RA paper (TeX) and the RA repository.
Restate the model and the algorithm in your own words.
List every equation and where it is implemented."
2. "Audit the code against the document. Report every place
where the code and the TeX disagree: scoring, bounds,
sampling, penalties. Classify each as a correctness risk
or a documentation gap."
3. "Propose a refined RA document + code pair in which the
two are consistent. Do not add any new economics yet."

- ▶ Prompt 1 tests comprehension; prompt 2 tests the seed; prompt 3 fixes the foundation while freezing the economics. The audit's findings were kept as a file.
- ▶ **The audit is the cheapest insurance in the project: an inconsistent seed propagates into every later version.**

Source: The instructor's research record: the project's opening prompts, typed as shown (RA: representative agent), described by role.

The Division-of-Labour Contract

the agent

drafts and reorganises the LaTeX
turns equations into vectorised code
proposes penalty and boundary patterns
summarises diagnostics and mismatches

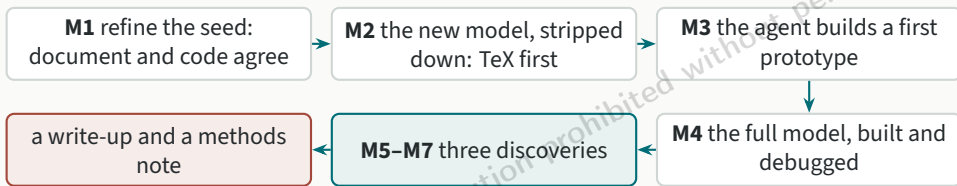
the economist

the economics of every formulation
which simplifications are acceptable
conceptual bugs: timing, direction
the benchmarks, and accepting results

- ▶ Written down at the start, before any code: who owns which kind of error.
- ▶ **The agent supplies throughput; the economist supplies judgement.**

Source: The instructor's research record: the project's division of labour, described by role; Miao & Feng, *AI for Economists*, ch. 15, §4.1.

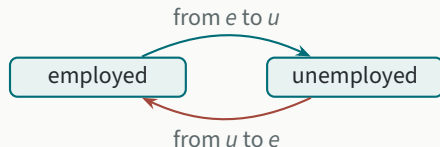
The Milestone Ladder, and Which Session Did What



- ▶ Separate sessions worked separate layers — model drafting, code review, debugging, documentation — all feeding one package.
- ▶ Each milestone was a validated artifact, kept; each failure of one version became explicit structure in the next.
- ▶ **The unit of progress is the validated milestone, not the week.**

Source: The instructor's research record, by role; Miao & Feng, *AI for Economists*, ch. 15, §4.1 and §4.3.

Words Before Code: Two Bugs No Compiler Catches



$$a_{t+1} = (a_t + y_t - c_t) / q_t$$

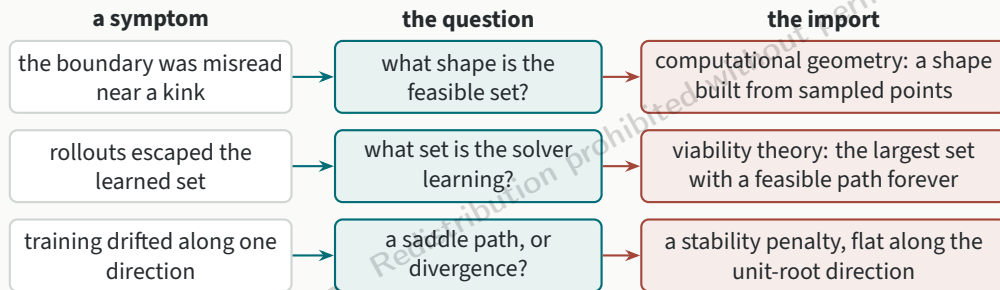
with c_t , not c_{t+1}

a assets, y income, c consumption, q
the bond price

- ▶ The agent drafted the stripped-down model's equations: syntactically fine, economically wrong in two places.
- ▶ The asset transition used next period's consumption; the budget constraint is a current-period restriction.
- ▶ The transition probabilities ran against the direction of movement between the employment states.
- ▶ **No compiler, test or residual catches a timing error. An economist reading the TeX does.**

Source: The instructor's research record: the stripped-down model, described by role.

Three Discoveries the Iteration Surfaced



- ▶ The second is the structure of self-generation in repeated games: the feasible set is the largest fixed point of a set operator.
- ▶ **Each came from iterating on why a run failed: the economist recognised the question; the agent carried the search.**

Source: The instructor's research record, described by role; Aubin (1991); Abreu, Pearce & Stacchetti (1990).

To Benefit the Most, You Need Four Things

a good seed

a working triple that points at the target

a clear vision

which problem is interesting, and why

good judgement

reject the plausible, wrong equation

orchestration

sessions on separate layers, one package

- ▶ What changed: the theory–code–debug loop compressed; documentation became executable; imports from other literatures got cheap.
- ▶ What did not is Deck 6.1's list: the AI never chose the question, never caught the timing on its own, never knew when a result was not credible.
- ▶ **The workflow is the product: it is the part you reuse on the next project.**

Source: The instructor's research record, by role; Miao & Feng, *AI for Economists*, ch. 15, §4.1–§4.3.

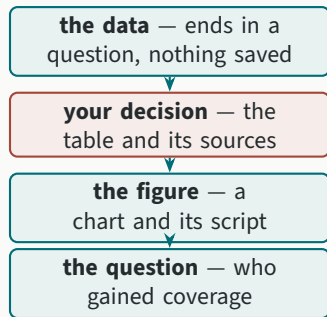
Session 6.3

The Checks That Stay Yours

What a clean run hides, how far the loop can run alone, and what to test before a number reaches the paper



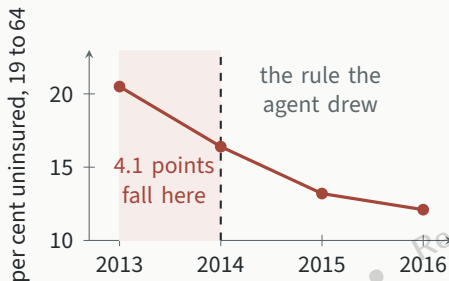
A Twenty-Two-Minute Task, in Four Prompts



- ▶ From an empty folder: four prompts, twenty-two minutes, recorded 13 September 2026 — uninsurance rates by age, 2008 to 2024, every source written down.
- ▶ The first ended with no data at all: the data service now wants a key issued to a person, so the agent stopped and asked.
- ▶ **The second prompt was not a correction. It was the judgement the agent lacked: one survey, and check the age groups before combining years.**

Source: Course-recorded agent run, 13 September 2026; U.S. Census Bureau, health-insurance historical tables (ACS).

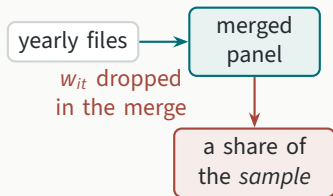
The Trap: An Instruction the Data Cannot Carry



- ▶ The fourth prompt asked for a January mark. Each point here is a one-year average sitting at its integer year.
- ▶ No position on this axis is a month, so the rule went at 2014 — the whole-year average, after the event.
- ▶ 4.1 of the 8.4-point fall, 48.8 per cent of it, ends up on the “before” side.
- ▶ **Every check passed. The instruction had no valid reading, and nothing in the loop says so.**

Source: Course-recorded agent run, 13 September 2026; U.S. Census Bureau, one-year estimates (ACS).

A Quiet Failure: The Lost Survey Weights



no error, no warning, every year wrong the same way ●

- ▶ The estimand is one line of arithmetic, the weighted uninsured rate

$$\hat{r}_t = \sum_i w_{it} \mathbf{1}\{\text{uninsured}\} / \sum_i w_{it}.$$

- ▶ Merging several annual files, an assistant lost the weight column: the pipeline reported the share of the *sample*, not the population rate.
- ▶ **Divergence announces itself; a pipeline error does not. This one surfaced at a group meeting.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4.4; AHRQ, Medical Expenditure Panel Survey documentation.

Passed Every Gate, and Still Wrong: Our Own Numbers

Session 2

| 0.4%



Session 4

| 1.5%

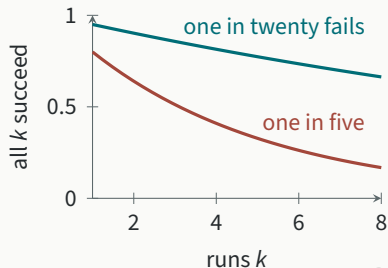


teal is the residual the check saw;
red is the error in aggregate capital

- ▶ Session 2, rung 1 from the default start: a held-out Euler residual of -2.38 on Judd's scale, the policy 25 per cent wrong, and capital at 0.06 against 28.35.
- ▶ Session 4, block D from a random start: a residual of -1.81 , capital 3.24 against the classical 5.64.
- ▶ **In both the machine check passed and the economics failed — which is why gate D reads capital within 5 per cent first, and the residual below -2.5 second.**

Source: Session 2's lab, block D (rung 1, default start); Session 4's lab, block D (random start) and its gate file.

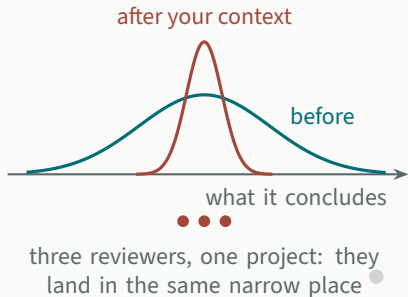
One Try Is Not a Measurement



- ▶ An agent left to run is a repeated trial: report the rate over k fresh runs, never the best run. pass^k , the chance all k succeed, falls fast even from high reliability.
- ▶ Published scores break two ways: leaked answers, and harness exploits that score well without solving anything — which is why no leaderboard number appears in this session.
- ▶ **Your own evaluation is a benchmark too: an agent that does nothing should score zero on it.**

Source: Yao et al. (2024), pass^k ; Miao & Feng, *AI for Economists*, ch. 15, §4.4.

The Context Dilemma, and the Prior Your Team Inherits



- ▶ Context acts as evidence: it narrows the output towards your framing, not the truth.
- ▶ Sharma et al.: sycophancy across frontier models — factual flip-flops, unwarranted concessions, matching the user's view.
- ▶ Perez et al.: it *scales with model size*; preference data reward the agreeable reply, so no “do not flatter me” brief line removes it.
- ▶ **Hint the theory “should” work and the model defends, not tests, the framing: keep one reader outside your context.**

Source: Xie et al., ICLR (2022); Ouyang et al. (2022); Sharma et al. (2023), arXiv:2310.13548; Perez et al. (2023), ACL Findings.

The Autonomy Slider, and Where Loops Converge

a machine decides

nobody but you

the tests pass

is the instrument
exogenous?

the benchmark
is matched

does the welfare
result hold?

the proof
checks in Lean

is the question
worth asking?

48 multiplications,
not more

is the result
believable?

left: the loop converges; right: it
drifts towards what it was last told

Source: Google DeepMind, AlphaEvolve (14 May 2025, a vendor account); Karpathy (2025); Miao & Feng, *AI for Economists*, ch. 15, §4.3.2; Deck 1.1's frame on the machine-checked proof.

- ▶ AlphaEvolve (DeepMind, May 2025): 4×4 complex matrices in 48 scalar multiplications, improving on Strassen (1969); a count decides, so the loop converged, as Deck 1.1's Navier–Stokes run did with a proof assistant.
- ▶ Set the autonomy slider per task: as far right as the criterion is decidable, and no further.
- ▶ Loop the decidable sub-tasks (tests, tuning, a benchmark it runs); keep the rest.
- ▶ **An agent that builds an agent stops where machines cannot decide right and wrong.**

Seven Tests Before It Goes into the Paper — and When Not to Delegate

test	the question
correctness	did it compute the object you meant; can you re-check magnitudes yourself
reproducibility	can a coauthor rerun a clean copy, with model, prompt, seed and data pinned
auditability	can a referee trace the commands, checks and sources behind it
privacy boundary	which files stayed local, and what content crossed the network
cost	what it cost in tokens and in money, including what a replicator will pay
speed	net of checking and repair, did it save time
skill investment	is your group building durable skill, or quietly losing it

- ▶ Keep a two-line edit or one-shot plot in the editor (no validation surface), and write the V0 gate before bringing in an agent: faster output is faster wrong.
- ▶ **Any “no” or “I do not know” means the output is not yet research-grade.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §5, the practice checklist, recast as tests of an output; the two boundary rules carried from the 2025 lecture on governance.

Five Habits, and What Deck 6.4 Adds

a system set up once

a brief, a team with a reviewer who did not build it, gates that refuse

restate the task
before any code

write the gate
before the run

grade the plan,
then approve

add one margin
at a time

keep one reader
outside your
context

- ▶ Each habit is cheap, and each is what one of this deck's two cases turned on.
- ▶ **Next, Deck 6.4: the one thing on your disk the agent still cannot read — your literature.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §4 and §5.

Your Turn: Lab Blocks B and F, and the Design Canvas

block B — recompute all 96 numbers of the recorded run, then find the axis



block F — a gate that fires, one that never fires, and one agent run again



the canvas — eight cells to fill before an editor opens

- ▶ Block B is Part D's twenty-two-minute task, put under audit: the arithmetic reproduces; the axis does not.
- ▶ Block F runs Part A's rules as code. Self-paced, after class, offline; about forty minutes for both.
- ▶ **Then fill the canvas for a project of your own, before you open an editor.**

Source: The session's lab, blocks B and F, and its design canvas.

Session 6.4

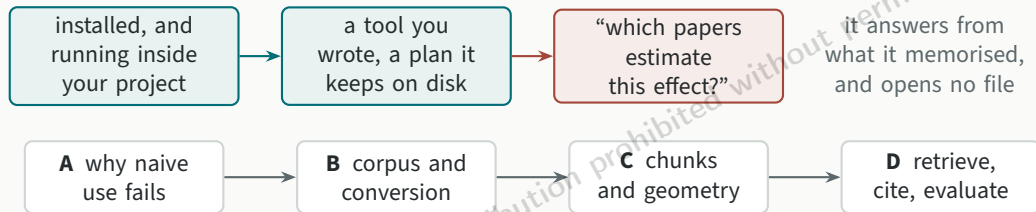
Build Your Own RAG: Give Your Agent a Library

Part four of five: why an agent cannot read your literature, what naive use costs, and a six-step recipe from a folder of PDFs to a cited answer — tied to Session 5 at each step



Zhigang Feng

Your Agent Runs Your Code, and Cannot Read Your Literature



- ▶ Decks 6.2 and 6.3 gave the agent hands: it runs commands, calls tools you wrote, and keeps a plan on disk.
- ▶ None of that put your literature in front of it. The shelf of papers on your own disk is still outside the loop.
- ▶ **Decks 6.2–6.3 closed the gap between advice and action. This deck closes the gap between action and evidence.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §1 and §2.1.

Session 6.4

Why Naive Use Fails

Three symptoms, two causes, the bill for pasting a paper, and three ways out



Three Symptoms of One Question

author, year and
title: all plausible

no such
paper

an argument, correct
somewhere else

a premise
is missing

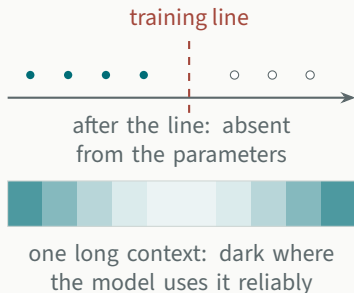
a fluent,
confident paragraph

no page,
no link

- ▶ Ask an unaugmented model which recent papers estimate your effect: three answers are possible.
- ▶ **Fabricated citation**: real scholars, a plausible year, a title in the right house style — and no such paper.
- ▶ **Misapplied method**: an argument right elsewhere, void here, and no warning given.
- ▶ **Confident and unsourced** is the third, and for academic writing it is a veto on its own.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.1.

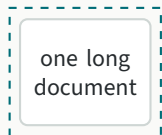
Two Causes: One “Does Not Have”, One “Cannot Use”



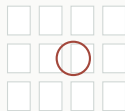
- ▶ **Knowledge cutoff** is a *does not have*: nothing past the training line is in the parameters.
- ▶ **Lost in the middle** is a *cannot use*: accuracy is highest at both ends of a long context and lowest in the middle.
- ▶ Two biases stack — recency, from causal attention, and primacy, from where instructions sit in the training data.
- ▶ **The constraint is effective attention, not capacity.**

Source: Liu et al. (2024), “Lost in the Middle”, *TACL* 12, 157–173.

Which Problems a Wider Window Actually Solves



capacity
binds: a wider
window helps

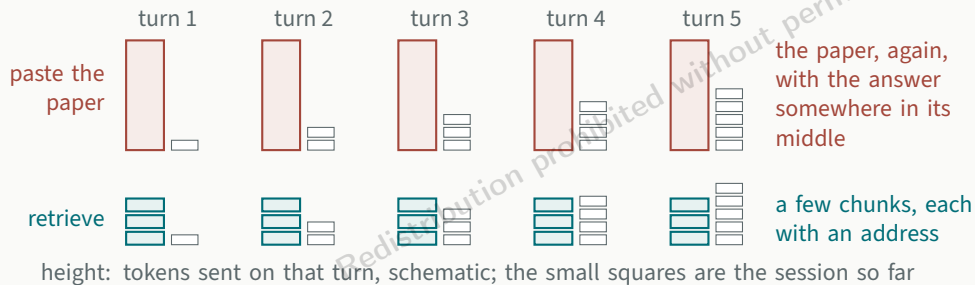


location and
provenance
bind: width
buys nothing

- ▶ Sort the information need into two kinds before choosing the tool.
- ▶ **Read one long document:** a working paper, a long policy file. Capacity binds, and a wider window genuinely helps.
- ▶ **Locate and cite:** a few passages somewhere in the whole corpus. No width helps; you need an address and a source.
- ▶ **If the task is both, split it** — and note that a wide context is paid for again at every question.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.1.

The Cost of Naive Use: Paste the Paper, Pay Every Turn



- ▶ Paste a paper once and every later question re-sends it, on any topic (Deck 6.2).
- ▶ The passage you need sits in the middle, where it is used least reliably (Deck 5.3; f3).
- ▶ **Retrieval sends only what the question needs, with each chunk's address: the bill stops growing with the paper.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.1–§2.2; Liu et al. (2024), *TACL* 12.

Three Ways to Give a Model Your Literature

paste it

the paper in the prompt,
re-sent every turn

context: this turn and
every turn after (f5)

index it

cut, embed, re-
trieve; cite the chunk

retrieval: an address
per passage, built once

send an agent

it searches the web and
writes a cited review

deep research: the
misses look like hits

- ▶ Rao et al. (2026), ten models and agents: 3–13 % of citation URLs are hallucinated, 5–18 % non-resolving. Onweller et al. (2026), fourteen models: link validity above 94 % and relevance above 80 %, yet only 39–77 % factual accuracy.
- ▶ **The misses look like hits:** a live link, a relevant page, a wrong sentence. Your own index cites or declines — this deck.
- ▶ Not on this list: fine-tuning changes how the model speaks, not what it can cite (f7).

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.2; Rao, Wong & Callison-Burch (2026), arXiv:2604.03173; Onweller et al. (2026), arXiv:2605.06635; both preprints, read 23 September 2026.

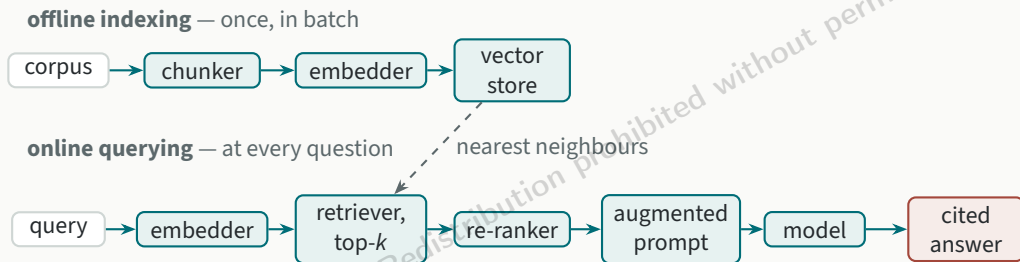
Fine-Tuning Versus Retrieval: Six Dimensions

dimension	fine-tuning	retrieval
compute	a training run, on GPUs	index once, then query
update delay	days: retrain and redeploy	minutes: add a document
attribution	none, by construction	built in: one tag per claim
style, format	strong: voice, terms, layout	weak: the base model writes
hallucination	limited: recall is parametric	stronger: held in the context
typical use	imitation, house terminology	live corpora, policy questions

- ▶ The table points one way for literature questions; the hybrid fine-tunes the embedder, not the writer.
- ▶ **Fine-tuning's comparative advantage is style, not knowledge.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.2, the six-dimensional trade-off.

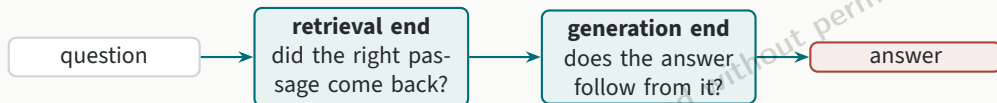
The Pipeline in One Picture



- ▶ **Offline indexing** runs once: chunk, embed, store the vectors and their metadata.
- ▶ **Online querying** runs per question: embed, retrieve, re-rank, inject, answer.
- ▶ **Index once, query many**: a fixed cost against a marginal cost near zero; each question ships 5 to 20 chunks, not the corpus.
- ▶ **Both chains use one embedder**: similarity exists only inside one space.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3; Lewis et al. (2020), *NeurIPS* 33, 9459–9474.

Score It Before You Build It: Two Ends, a Score for Each



a good score at one end and a bad one at the other is not a good system — it is a system that fails in a way you have not measured

- ▶ An economist has met this shape before, and it runs the other way. A doubly robust estimator is safe if *either* the outcome model or the propensity model is right; here a good score at one end buys nothing if the other fails.
- ▶ Write the score first, with questions the corpus cannot answer: RAG's V0 gate (Deck 6.3).
- ▶ **Both ends must pass. They fail for different reasons and are repaired in different places.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.4.

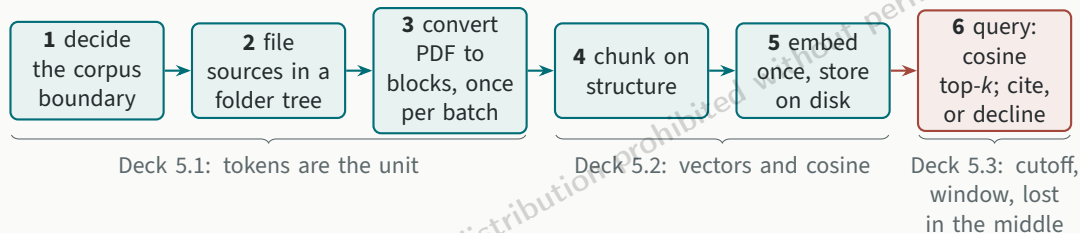
Session 6.4

Stages 1–2: Corpus and Conversion

Deck 5.1 made tokens the unit: the model reads text, so the first jobs are deciding which text, and turning files into it



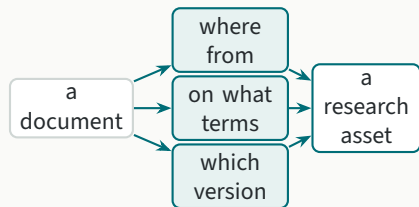
The Recipe, in Six Steps



- **Steps 2 to 6 are engineering you can copy. Step 1 is research only you can do.**
- f8 drew the pipeline on paper; this part walks one in daily use — 162 documents, 74,557 chunks, one index on disk — and rung two is the control condition Deck 6.5 has to beat.
- **The goal is not to admire it. It is to leave with the recipe for your own corpus.**

Source: Instructor's working retrieval infrastructure, read from source on 23 September 2026; Miao & Feng, *AI for Economists*, ch. 15, §2.5.

The Corpus Is a Research-Design Choice

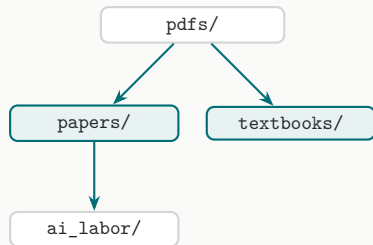


any one missing: a pile of files, not a corpus

- ▶ The corpus is the system's whole knowledge: what is left out cannot be retrieved.
- ▶ Ready-made, one per field: a central bank's archive, a filings database, a working-paper archive.
- ▶ **A corpus with no licence is not a research asset:** no provenance, no terms, no frozen version, nothing a referee can re-run.
- ▶ This session: 245 public-domain statements, 1994-02-04 to 2026-06-17. The worked system: 162 documents, 88 of them papers.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3 and §2.5; this session's frozen corpus manifest; the working system read from source on 23 September 2026.

The Folder Tree Is the Metadata; the Hash Prevents Repeat Work



a document's path *is* its topic

- ▶ There is no separate topic field to maintain. Filing a paper in a folder is how it gets classified.
- ▶ A manifest row carries a content hash, so a rerun passes over converted sources; each numbered stage reruns alone.
- ▶ **The cheapest metadata schema is the one you cannot forget to update.**

Source: Instructor's working retrieval infrastructure, read from source on 23 September 2026.

Stage 2: A PDF Becomes Readable Context

```
export LC_ALL=en_US.UTF-8 # CJK file names, spaces in paths
STAGE="$(mktemp -d)"; for f in "$IN"/*.pdf; do n="$(basename "$f" .pdf)"
[ -d "$OUT/$n" ] || ln -s "$(readlink -f "$f")" "$STAGE/"; done # unconverted
mineru -p "$STAGE" -o "$OUT" -b vlm-auto-engine # the converter runs ONCE per batch
ls "$OUT" | wc -l; wc -w "$OUT"/*/vlm/*.md # files kept? words kept?
```

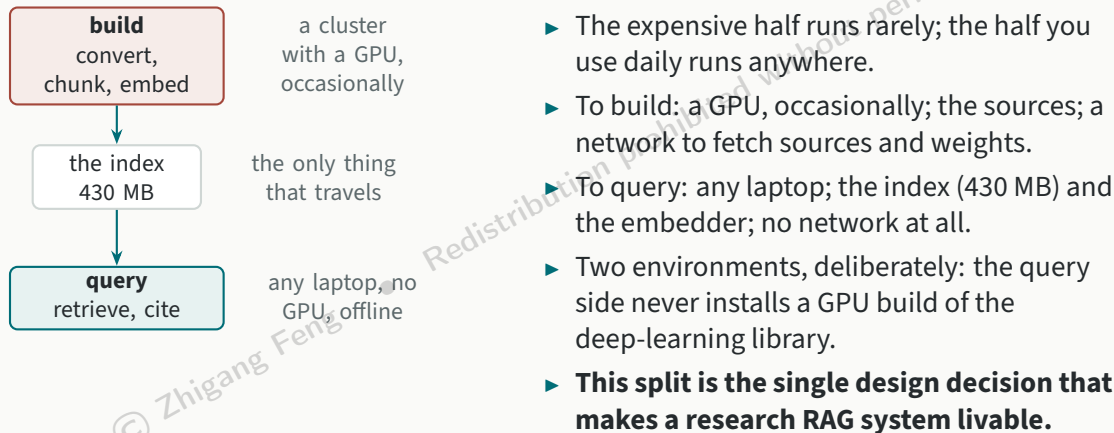
a 54-page paper, one A100 GPU

engine start-up, model load	110 s , <i>fixed</i>
inference, all 54 pages	0.8 s <i>a page</i> , <i>marginal</i>

- Conversion has **a fixed and a marginal cost**: called once *per file*, thirty PDFs cost fifty-five minutes of model loading around twenty minutes of work. Stage the batch, call it once.
- Normalise early: every format becomes one block record. The silent failure in the logs: a concurrency default deadlocks the engine.

Source: Course conversion benchmark, recorded July 2026 (the benchmark folder no longer exists; the three numbers are carried from the legacy frame of that date).

The Design Rule: Build on a GPU, Query Anywhere



Source: Instructor's working retrieval infrastructure, read from source on 23 September 2026.

Session 6.4

Stages 3–4: Chunks and Geometry

Deck 5.2's vectors and cosine: cut the text into citable units, then place each one where nearness means meaning



Why Chunk at All; Five Strategies, Each More Respectful of Structure

“the effect is significantly negative”

which effect?
which sample?

one block,
three topics

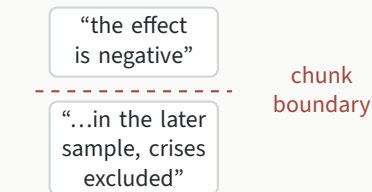
precision
diluted,
the prompt
crowded

the unit of retrieval is
the unit of evidence

- ▶ **Chunking** cuts the corpus into the units retrieval returns. Too small, no context; too large, several subjects and a crowded prompt.
- ▶ **Fixed length, sliding window, recursive character**: each breaks at a more natural boundary.
- ▶ **Semantic**: cut where similarity between neighbours drops.
- ▶ **Document-aware**: headings, tables, metadata; on economic text, almost always worth its cost.
- ▶ **Cut at the unit you would be willing to cite.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3.

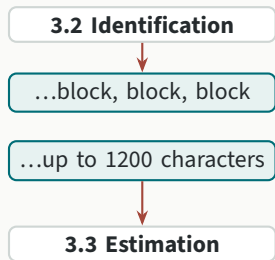
The Economist's Trap: A Claim Cut From Its Condition



- ▶ Economic text is already structured: a statement has an action, an assessment and a vote.
- ▶ The trap is cutting a claim away from the condition it holds under: the sample, the specification, the date.
- ▶ Document-aware chunking keeps the unit whole and puts the heading in the metadata.
- ▶ **A chunk that drops its conditions is worse than none:** it reads as general and is not.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3, on structured economic corpora.

Headings Are Hard Boundaries, and What One Chunk Carries

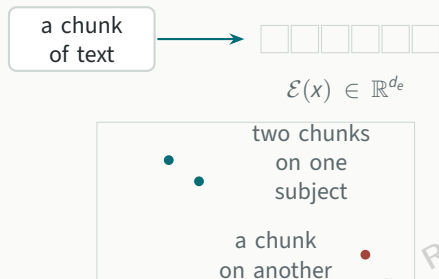


a new chunk
starts here

- ▶ Blocks accumulate to about 1200 characters, with a 150-character overlap and a 200-character floor.
- ▶ The overlap is whole blocks walked backwards: it never cuts a sentence, and *never* crosses a heading.
- ▶ Each chunk carries a **locator**, a page range or a section path: that is why an answer can cite.
- ▶ **A chunk that spans two sections is a chunk that answers neither question.**

Source: Instructor's working retrieval infrastructure, read from source on 23 September 2026.

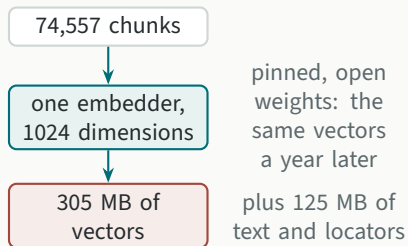
From Chunks to Geometry



- ▶ **Document embedding:** one vector for a whole chunk, $\mathcal{E}(x) \in \mathbb{R}^{d_e}$, with d_e usually between 384 and 4096.
- ▶ Deck 5.2 drew this geometry: nearness stands for closeness in meaning.
- ▶ So do its warnings: two filings from different industries look alike when the boilerplate does.
- ▶ **You do not implement this:** call an encoder, feed the chunks in batches, keep the vectors.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3; Deck 5.2 of this course.

Choosing an Embedder; One Bilingual Model, Vectors on Disk



- ▶ Choose on four things: the dimension you can store, the corpus's languages, domain fit, and whether the weights are yours to pin.
- ▶ One model embeds corpus and query alike; change the encoder and the index is stale, not slightly wrong.
- ▶ Bilingual corpus — 158 English documents, 4 Chinese — so the embedder is too.
- ▶ **Open weights are a replication requirement:** a hosted endpoint can be upgraded under you; a pinned version can be re-run. The build refuses vectors from another model.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3; the working system read from source on 23 September 2026.

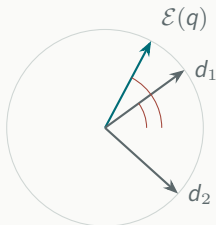
Session 6.4

Stage 5: Retrieve, Cite, Evaluate

Deck 5.3's three limits — the cutoff, the window, lost in the middle — are what the last step is built around



Similarity Search, and What a Vector Store Is For



cosine compares angles,
so chunk length drops out

- ▶ Compare query and chunk by the angle between their vectors:

$$\text{sim}(q, d_i) = \frac{\mathcal{E}(q)^\top \mathcal{E}(d_i)}{\|\mathcal{E}(q)\| \|\mathcal{E}(d_i)\|}.$$

- ▶ **Top-*k***: the retriever returns the 5 to 20 nearest chunks; the worked system returns 8. A **vector store** holds the vectors and the metadata and answers that query fast.
- ▶ **Nothing here understands the question**; it measures angles.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3; the working system read from source on 23 September 2026.

Two Lists, One Ranking: Keywords and Rank Fusion

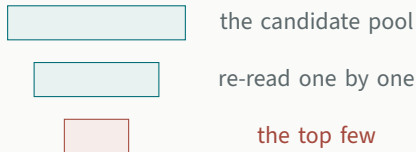
dense	sparse	fused
A	D	A
B	A	D
C	E	B

A tops one list and is second in the other, so it wins the merge

- ▶ Dense retrieval can walk past an exact string; **BM25** counts the literal term. Two lists, two kinds of score, in different units.
- ▶ So merge by *rank*:
$$\text{RRF}(d) = \sum_{r \in R} 1/(\kappa + r(d)), \text{ with } \kappa = 60.$$
- ▶ The paper, and the lab's code, call it k ; κ here keeps it apart from top- k .
- ▶ Agreement across lists beats one high rank; low-ranked chunks still count.
- ▶ **Fixed in a pilot and never re-tuned:** a default, not a finding.

Source: Cormack, Clarke & Büttcher, *Proceedings of SIGIR '09* (2009), 758–759.

Re-Ranking: A Cheap Screen, Then an Expensive Read



editors screen, referees
read: the expensive step
never runs on everything

- ▶ The first pass is fast and approximate: one stored vector per chunk against one vector for the query.
- ▶ **Re-ranking** hands the shortlist to a slower model that reads query and chunk together, then keeps the best few.
- ▶ **The screening step decides what the careful step can ever see:** a chunk missed in the first pass is not recoverable later.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3, on coarse screening and fine ranking.

Stage 5: Retrieve, Then Cite — the Augmented Prompt Has Four Parts

system: answer from the context only; tag every claim; if it is not there, say so

context: the top 8 chunks, each with identifier, title, page range and score

question

format: how to cite, how long

- ▶ The query is embedded by the same model, scored by cosine against every vector; the top 8 come back. A topic filter applies *after* ranking, so a narrow literature never starves the result.
- ▶ **Grounding** is the system line: answer from the context, fill no gap from memory; the template turns evidence into an answer.
- ▶ The only place a generative model is used. **Everything upstream exists to put good evidence in front of it.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3; the working system read from source on 23 September 2026.

The Citation and the Refusal, and a Worked Answer

Q. What did the Committee say at that meeting, and what did the ten-year yield do next?

A. one clause [doc-07]; a second clause [doc-12].

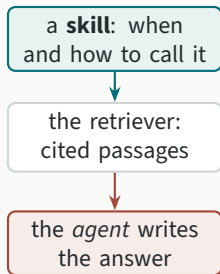
Q. And fiscal policy? **A.** Not in the retrieved context.

each clause carries the identifier of the chunk it came from, assigned at chunking time

- ▶ One stable identifier per chunk: required by the template, resolved back to the page.
- ▶ In the lab the refusal is a number: the best evidence score among the top 5 chunks against a threshold of 0.218; 4 of 6 unanswerable questions are turned away.
- ▶ **“Not in the context” is the system working:** a refusal costs one question, a fabrication costs the paper.
- ▶ **A tag proves provenance, not correctness:** whether the right chunk came back is measured, not assumed — the last part of this deck.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3 and §2.5; the session’s lab protocol.

Making It a Tool Your Agent Can Reach



a second skill
at ingest: read
the front page,
fill in author
and year

- ▶ Deck 6.2 defined a skill as a folder the model opens when needed. This is one, pointed at a real index.
- ▶ The retriever never calls a language model: it prints evidence and an instruction — answer only from this, cite every claim, say so if the evidence is weak.
- ▶ **The generator is the agent you already have: the library is one more tool.**
- ▶ Four seams, cut in advance: a format is one converter, a chunker one strategy, a store one interface, settings one file.

Source: Instructor's working retrieval infrastructure, read from source on 23 September 2026.

Faithfulness: The One That Matters Most

- ▶ Break the answer into claims. For each, ask whether the retrieved passages actually support it.
- ▶ A fluent answer that adds one unsupported sentence is more dangerous than an answer that refuses.
- ▶ **This is the score that stands between a citation and a fabricated one.**
- ▶ In the lab you mark one answer by hand:
 $F = \#\{j : C \models c_j\} / m$, the share of its atomic claims that the context supports.

claim 1: supported

claim 2: supported

claim 3: nowhere
in the passages

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.4; the course lab's gold set.

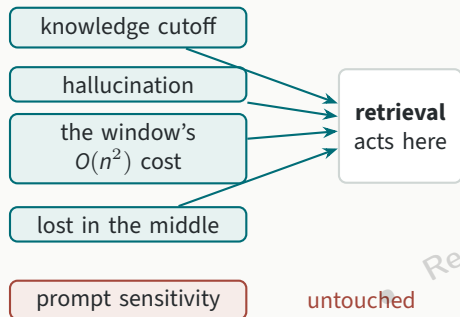
Five Ways It Breaks, and Where Each Sits

What you see	Where it sits	What repairs it
nothing relevant comes back	the index	the corpus boundary
the right passage ranks fourth	the ranking	a second scorer, or re-ranking
a claim lost the condition it held under	the chunker	boundaries that respect structure
a confident answer about a year you never indexed	the query	a refusal rule
a confident answer from the wrong meeting	an ambiguous query	ask it back before answering

The lab produces three on purpose: a stale index, a cut inside the decision sentence, an ambiguous question answered from the wrong meeting. **The symptoms read alike from outside. Diagnose by asking which stage could have produced this one.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.4.

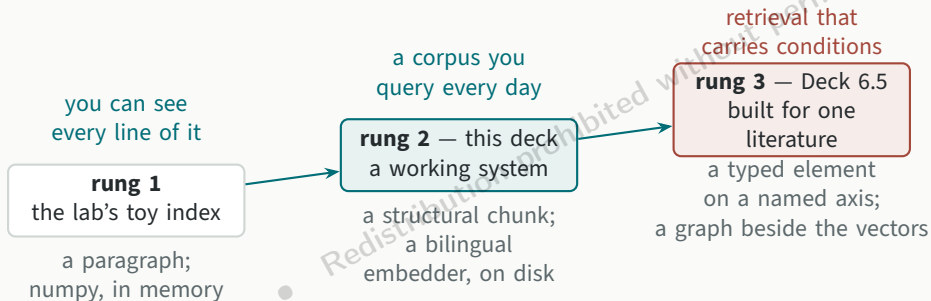
The Five Limits Session 5 Left Standing, Now Four



- ▶ Deck 5.3 closed on five limits of a pretrained model, all architectural: a larger model does not remove them.
- ▶ The six steps repair four by one move: the knowledge left the parameters for an index you can rebuild today.
- ▶ **The fifth is untouched.** Prompt sensitivity belongs to the prompt, not the evidence.
- ▶ Two limits remain: the recipe refuses by a sentence in its prompt (the lab, by a threshold), and a dense vector cannot tell you when it is wrong.

Source: Miao & Feng, *AI for Economists*, ch. 13, §6.1, and ch. 15, §2.5.

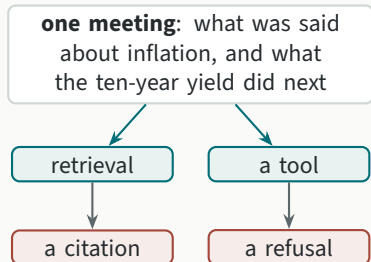
Three Rungs: Toy, Working, Research



- ▶ This deck built rung two on a corpus of your own; Deck 6.5 is GRAM, built for one literature.
- ▶ **Each rung answers what the rung below answers badly, and costs more to keep standing. Climb when the rung below is boring.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3 and §2.5.

Your Turn: Lab Blocks C, D and E



drop any one and the task is still answerable — fluently, and wrongly

- ▶ **C**, build the index: 245 statements, chunks that name their statement, two lists fused by rank.
- ▶ **D**, answer or refuse: a threshold fitted on the graded set, and three failures produced on purpose.
- ▶ **E**, faithfulness by hand: split one answer into claims and mark each one supported or not.
- ▶ **Self-paced, after class, offline: a CPU, no account, fixed seeds.**

Source: The session's lab, blocks C to E; Session 2's frozen yield panel.

Session 6.5

GRAM: A Retrieval System Built for One Literature — and What Agents Do to Research

Part five of five: rung three, how it was built and how the method transfers; then the close: the two dangers answered, the evidence on the researcher, and the economics of an agent

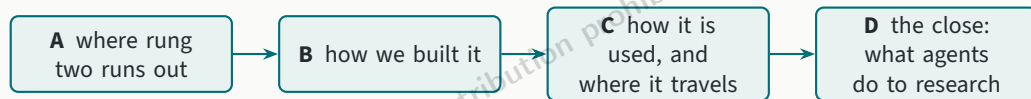


Zhigang Feng

This deck describes a system that is work in progress by Zhigang Feng, Jianjun Miao, Haoxuan Xie, Yaolong Zhong, and Yingli Zhou (2026). Its evaluation is a design; no result is reported here.

The Query This System Was Built For

“Life cycle with overlapping generations, idiosyncratic and aggregate shocks, incomplete intergenerational asset markets — which solver class is admissible for the joint stationary distribution in recursive equilibrium?”



- ▶ Every noun in it is a *condition*. The question is not what a paper is about; it is which method stays valid once those conditions hold together.
- ▶ A flat index matches the vocabulary and returns the famous paper that shares it — a different question, answered confidently.
- ▶ **Keep this sentence in view. Every design choice in this deck is an answer to it.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

The Pilot, and What These Pages Add

the newest frontier working papers

the last decade of journal papers

the works that fixed
mechanism and method

the founding early papers

one subfield: overlapping generations
and the life cycle

- ▶ **GRAM** — Generative Reasoning and Augmented Modeling — is a retrieval system for one literature in quantitative macroeconomics.
- ▶ The pilot **built 133 papers end to end**: the number carried through every stage into the index, not the number selected for it.
- ▶ The book already publishes the layered curation, the typed basis and the evaluation design; these pages add the reasoning and the engineering, and no result.
- ▶ **One subfield, deliberately. The claim is about a method, not about coverage.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.5; Feng, Miao, Xie, Zhong & Zhou (2026), work in progress; pilot build count.

Session 6.5

Where Rung Two Runs Out

A question about admissibility, and why a flat index answers it by prominence



A Question Flat Passages Answer Badly

a passage: "...aggregate shocks
...recursive equilibrium ..."

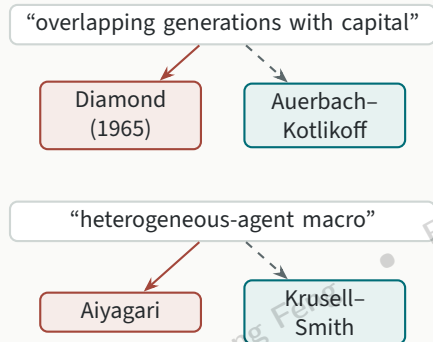


what it does not say: what
enters which constraint, and
what clears in which market

- ▶ Rung two's unit is prose: it carries a mechanism's words without the mechanism. Two papers that impose a borrowing constraint differently write nearly the same paragraph.
- ▶ A question about a *relationship* is not one about a topic. No passage holds the relation between two papers; it lives in the edge between them.
- ▶ **A flat index ranks on wording, and admissibility is not a property of wording.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

Wrong by Prominence



Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

- ▶ Embedding retrieval routes a query to the *most prominent* paper carrying its vocabulary, fitted to the asked-for conditions or not.
- ▶ In red, what comes back. In teal, the paper whose conditions actually match.
- ▶ **The vocabulary is present; the applicability filter is absent.**

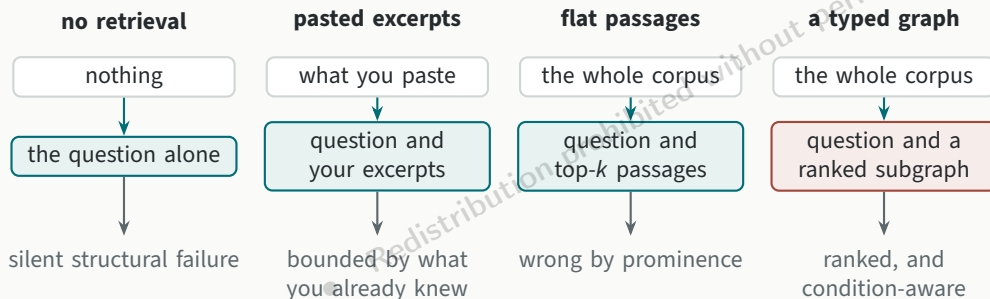
Admissibility Is Conditional

- ▶ Add aggregate risk to a life-cycle model and a solver class that was admissible stops being admissible; another one becomes so.
- ▶ The dependence runs from assumptions to methods, and neither paper's prose states it.
- ▶ **A vector of word co-occurrences cannot encode a dependency. A walk over typed objects can.**

Condition-aware retrieval returns a method *together with the conditions under which it holds*, rather than a passage that reads correctly and quietly drops its range of validity.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.5.

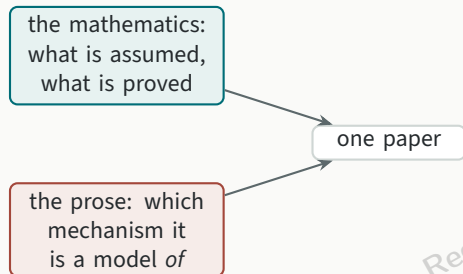
Four Ways to Give a Model Context, on One Question



- ▶ The generator is the same in all four. Only what reaches it changes.
- ▶ **These name each mechanism's failure mode on one kind of question. They are not a ranking of architectures.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

Why Economics Is Special: A Paper Is Mathematics *and* Prose



- ▶ Retrieval over the mathematics alone finds formally similar objects that model different things.
- ▶ Retrieval over the prose alone finds papers on the same topic under assumptions that do not hold.
- ▶ Economics needs both at once, which is why neither a mathematics search engine nor a general document index is the right starting point.
- ▶ **The index has to hold the structure, not a rendering of it.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

Session 6.5

How We Built It

Three models, a curated corpus, five stages, and a basis with names on its axes



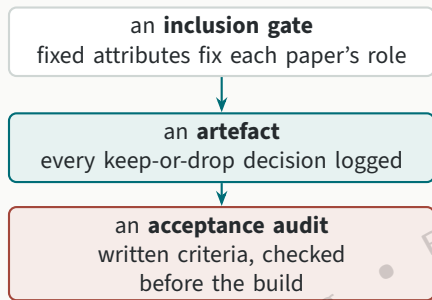
Three Models, Three Roles

role	modality	maps	what it is asked to do
reading	vision– language	page images to mark-up	rebuild the page: layout, reading order, tables, equations
reasoning	text to text	text to a typed record	resolve citations, and place the paper on the axes
embedding	text to vector	text to one space	put every element, and the question, in that space

- ▶ The role is named, not the product: each is replaceable, and the pipeline has to survive the replacement.
- ▶ **They never substitute for one another. A system that uses one model for every job cannot say which part failed.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

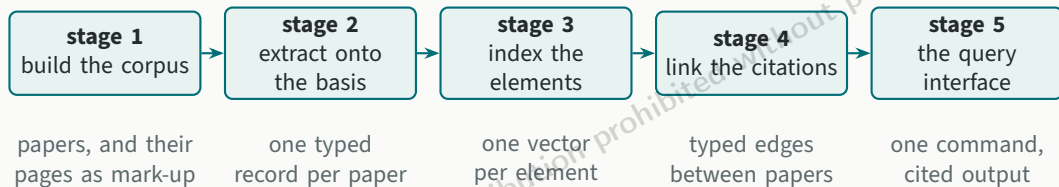
Corpus Construction Is Engineering: Tiers and Their Pressures



- ▶ A reading list is a personal judgement. A corpus is an artefact someone else has to be able to rebuild.
- ▶ Tiers settle three pressures, one layer at a time: coverage against focus, before against after the cut-off, and citation connectivity.
- ▶ **A retrieval system is only as good as the documents in its index, and that sentence is an instruction.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

The Five Build Stages

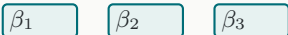


- ▶ No stage imports another: each reads the previous stage's files and writes its own, so any stage can be re-run alone.
- ▶ **Stages 1 and 2 are research; 3 to 5 are engineering. Keeping them apart is what lets you say which one failed.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

Why a Named Basis, and Not a Statistical One

named coordinates



each coordinate means something, so two papers compare one axis at a time

anonymous coordinates

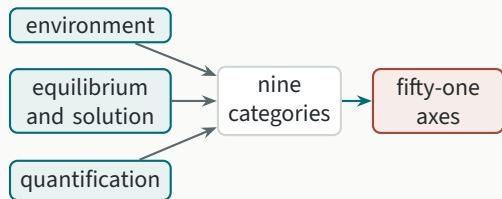


a direction that predicts well and explains nothing

- ▶ The analogy is a factor model: loadings on a few *named* factors,
$$R_i = \alpha_i + \beta_i^\top F + \varepsilon_i.$$
- ▶ A dense embedding gives a paper 4096 coordinates — the principal components. The fifty-one named axes are the factors.
- ▶ **You can ask which axis two papers differ on. You cannot ask that of coordinate 2,317.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress; the analogy is to named-factor asset pricing.

The Basis: Three Blocks, Nine Categories, Fifty-One Axes



preferences, technology, market structure,
solution algorithm, welfare measure, ...

- ▶ three blocks, capitalised — the environment, equilibrium and solution, quantification — and beneath them nine categories and fifty-one axes.
- ▶ Seven typed edges say how the pieces relate: what enters an objective, what enters a constraint, what clears in a market, what derives what.
- ▶ **The schema is the research. Extraction is the mechanical pass you automate.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

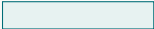
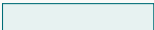
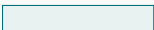
One Paper, One Provenanced Record

```
{ "axis": "borrowing_constraint",  
  "canonical_name":  
    "natural debt limit",  
  "description":  
    "the tightest limit that  
    never binds in autarky",  
  "source_lines": [412, 418],  
  "edges": [  
    { "type": "enters_constraint",  
      "target": "household_problem" } ] }
```

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

- ▶ One record per axis the paper touches, each carrying the lines it was read from.
- ▶ The canonical name is where papers merge — at the axis, never at paper-level similarity.
- ▶ **Provenance sits inside the retrieval unit, so every element returned can point at the sentence it came from.**

An Empty Axis Is Information

preferences	
technology	
market structure	
solution algorithm	

the blank row is a finding: this model has no such component

- ▶ A missing coordinate is a zero, not a hole in the data. It says the model contains no such ingredient.
- ▶ That is readable only because the basis is fixed in advance: with a schema invented per paper there is nothing to be absent from.
- ▶ **An absence you can query is a design choice. An absence you cannot see is a defect.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.5.

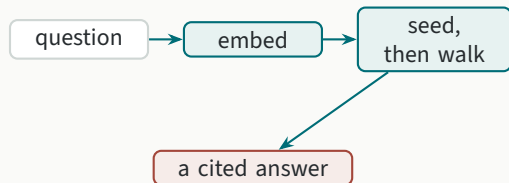
Session 6.5

How It Is Used, and Where It Travels

One command, a walk over a typed graph, and an evaluation that has not been run



At Query Time: Seed, Then Walk

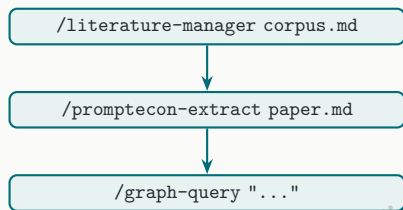


read-only and deterministic: the same question assembles the same context

- ▶ The model that indexed the elements also reads the question, so both live in one space.
- ▶ The nearest elements are not the answer; they seed a walk over the typed graph.
- ▶ The context policy is three numbers: eight seeds, then six element and four citation chunks.
- ▶ **The answer is written from the assembled block alone: cite it, or decline.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

The Interface Is a Slash Command



a verified manifest: what is in the corpus, and where each file came from

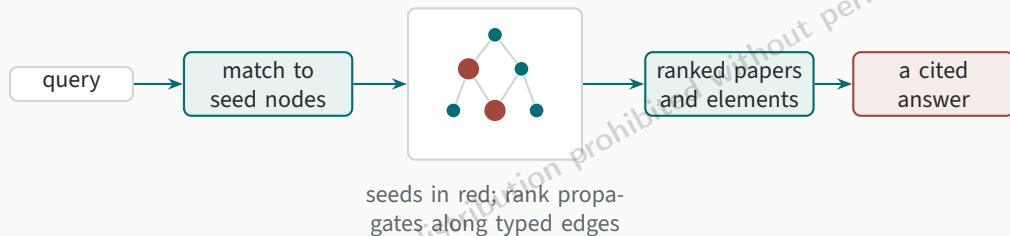
one typed record for that paper, placed on the axes

ranked papers, elements and citations — cited, or a refusal

- ▶ A skill is a folder the model opens when it is needed (Deck 6.2). These are three of them, in the order a researcher uses them.
- ▶ **The interface is part of the deliverable. A retrieval system you have to remember how to invoke is one nobody uses.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

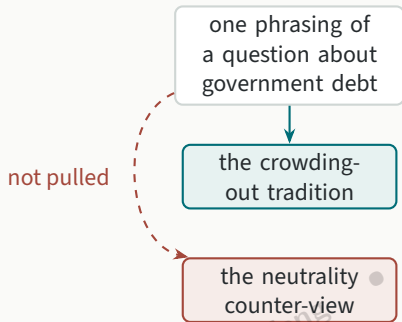
Graph Retrieval: Seeds, Propagation, Ranked Papers



- ▶ Far from the seeds, a paper never reaches the model: filtering precedes generation.
- ▶ End to end: embed, seed, propagate, assemble ten chunks, answer or decline.
- ▶ **A walk can express “reachable from these conditions”. A nearest-neighbour lookup cannot.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

Retrieval Honesty: The Counter-View That Did Not Surface

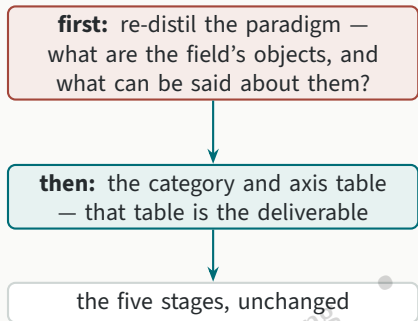


- ▶ In one demonstration the Ricardian-equivalence counter-view never surfaced, although the paper making it sits in the corpus.
- ▶ An answer is grounded in whatever retrieval pulled, so it is only as balanced as the phrasing that pulled it.

The assembled context must be printable. One-sidedness is visible there and nowhere else — never in the prose of the answer.

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress; a demonstration, not a measurement.

Reusing the Recipe in Another Field



- ▶ The engineering transfers unchanged. The basis does not: fifty-one axes for structural macroeconomics say nothing about an empirical literature.
- ▶ Re-deriving the paradigm is months of reading, and it is the step people try hardest to automate.
- ▶ **The table of categories and axes is the contribution. The pipeline around it is plumbing.**

Source: Feng, Miao, Xie, Zhong & Zhou (2026), work in progress.

From Fifty-One Axes to Your Field: A Worked Transfer

object	category	axes
identification strategy	the research design	variation; comparison group; timing
data	source and unit	survey or register; unit; years
population	treated, and compared	age band; income band; jurisdiction
estimand	the parameter	average or local effect; which margin
outcome	what is measured	coverage; utilisation; health; finance

- ▶ The previous frame's three steps, run once on the empirical literature on health-insurance expansions: five objects, none a preference or a technology.
- ▶ Each object becomes a category; its axes are the values a paper can take, fixed before reading, so an empty axis still reads as information.
- ▶ **Stages and models unchanged. The table is the transfer, and the economist writes it.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.5; Feng, Miao, Xie, Zhong & Zhou (2026), work in progress; axes illustrative.

The Evaluation Is a Design. Nothing Is Reported Yet.

1 the model alone, no retrieval

?

2 flat passages, one
dense bilingual embedder

?

3 typed axes and
a citation graph

?

one generator across all three: the
manipulated variable is retrieval

- ▶ **Condition 2 is the system Deck 6.4 built** — the control condition rung three is measured against.
- ▶ Three questions: is the extraction faithful; which component earns any gain; does typed retrieval beat flat passages.
- ▶ Two tasks: an expert panel, and generating an algorithm and its code.
- ▶ **No condition has been scored. Nothing in this deck is a result.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.5; Feng, Miao, Xie, Zhong & Zhou (2026), work in progress — design only.

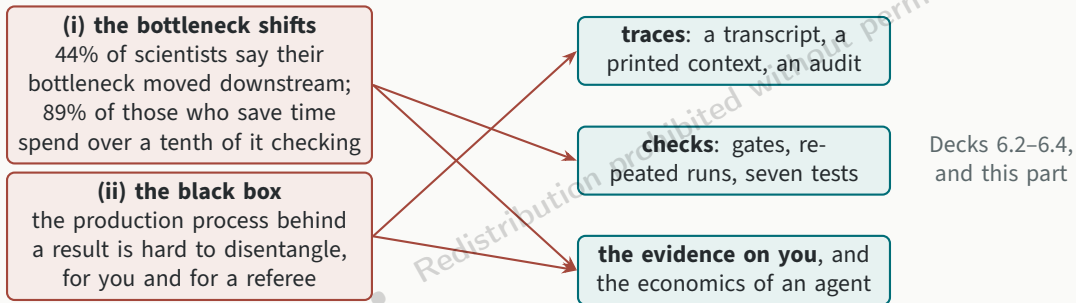
Session 6.5

Closing: What Agents Do to Research, and to You

The two dangers answered, the evidence on the researcher, and the economics of an agent



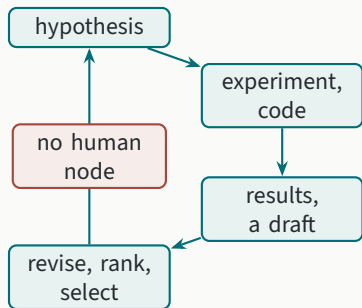
Two Dangers, Answered



- ▶ Deck 6.1 named both dangers and the survey behind (i). The answer to both is a contract kept in files: spec, trace, gate.
- ▶ **This is the part where the session stops adding capability and starts pricing it.**

Source: Codreanu, Imas, Mateos-Garcia et al. (2026), *AI in Science: Early Insights*, pp. 5, 25–26 (read 23 Sep 2026); Deck 6.1's closing frames; Holmström (1979).

Closing the Loop, and Where It Stops

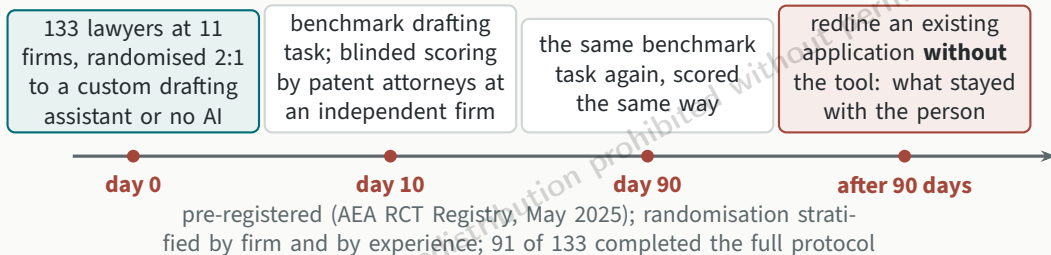


converges where the
criterion is auto-decidable
and iteration is cheap

- ▶ Lu et al. (2026, *Nature*): an end-to-end system's workshop paper scored 6, 7 and 6, above the bar, and was withdrawn as agreed; its failures included hallucinated citations and wrong implementations.
- ▶ Where a published table decides, loops converge first: the best agents match a reported sign over 85% of the time (Kohler et al., 2026).
- ▶ Scheduled, unattended runs remove the turn in which you would have noticed.
- ▶ **Nothing inside the loop knows whether the question was worth asking.**

Source: Lu et al. (2026), *Nature* 651: 914–919, 25 Mar 2026; Kohler et al. (2026), arXiv:2604.21965 v1, preprint; Anthropic, routines in Claude Code, 14 Apr 2026; read 23 Sep 2026.

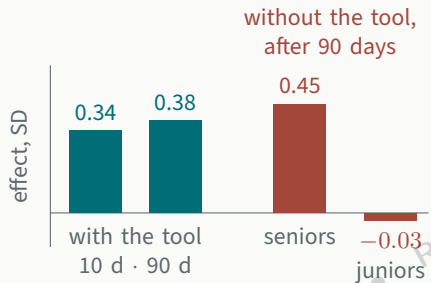
Does AI Assistance Enhance or Erode Expertise? The Design



- ▶ A three-month field experiment in patent drafting, at practising firms, with a rubric covering enforceability, accuracy, strategic ambiguity, completeness and clarity.
- ▶ **The design answers two questions at once: what the tool adds while you use it, and what remains when it is taken away.**

Source: Autor, Rodchenko, Martin, Iscenko, Strand, Pearl & Ferere (2026), NBER WP 35720, Sept 2026, pp. 1–3 (read 23 Sep 2026; not peer-reviewed).

Seniors Kept the Gain; Juniors Split



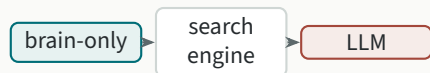
“Foundational expertise may be a prerequisite for extracting durable skill from AI-assisted practice.”

- ▶ With the tool: +0.34 SD at 10 days, +0.38 SD at 90, larger for juniors. Without it: +0.32 SD overall, all of it seniors (+0.45 SD); juniors -0.03 on average and *bifurcated* — fewer mediocre scores, more poor ones, more good ones.
- ▶ The counterpoint to Deck 5.5’s “cognitive levelling” (Noy & Zhang; Brynjolfsson, Li & Raymond): it holds only while the tool is in hand.
- ▶ **The largest gains accrued to the lawyers who retained the least.**

Source: Autor et al. (2026), NBER WP 35720, pp. 2, 18, 23–24, 38 (read 23 Sep 2026); effect sizes in SD, not peer-reviewed.

What Automation Does to the Researcher: Cognitive Debt, Hedged

sessions 1–3: connectivity,
strongest to weakest



“cognitive activity scaled down
in relation to external tool use”

session 4:
LLM →
brain-only

session 4:
brain-only
→ LLM

the crossover: LLM-to-brain showed
reduced alpha and beta connectivity

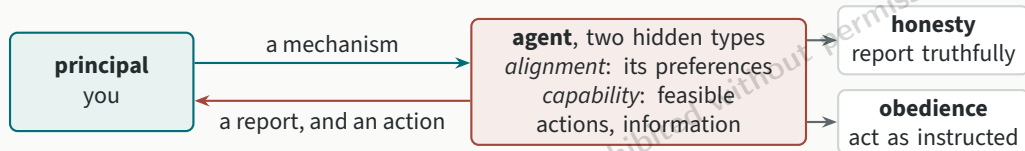
reliable tools get checked less

Preprint, under review (arXiv v2, 31 Dec 2025); $n = 54$, 18 in session 4. Gerlich (2025): AI-tool use and critical thinking, $r = -0.68$, mediated by cognitive offloading — a correlation in a UK survey ($n = 666$), which the book hedges the same way. Neither is a causal estimate.

- ▶ In session 1, 83.3% of the LLM group (15 of 18) could not quote a sentence of the essay just handed in; by session 2 the share had fallen.
- ▶ **Thin evidence, consistent direction: what you hand over, you stop rehearsing.**

Source: Kosmyna et al. (2025), arXiv:2506.08872 v2, pp. 31, 36; Gerlich (2025), *Societies* 15(1):6; Parasuraman & Manzey (2010); Miao & Feng, ch. 15, §4.4.3; read 23 Sep 2026.

Mechanism Design When Alignment and Capability Are Both Unknown



one-sided imitation: “capabilities can be concealed but not counterfeited” $\Rightarrow$ a revelation principle, and implementable policies characterised by nested cyclical monotonicity

- ▶ Deck 1.1’s principal could not observe effort; this agent hides what it wants and what it can do, so a mechanism “must incentivize both honesty and obedience”.
- ▶ A weak agent cannot fake strength, a strong one can play weak: this asymmetry makes it tractable; eliciting higher-order beliefs can discipline several agents at once.
- ▶ **Deck 6.1’s two dangers in one contract: unseen effort, unseen ability.**

Source: Bergemann, Koh & Morris (2026), “Mechanism Design for Alignment and Control”, arXiv:2609.01595 v1, 1 Sep 2026, abstract (body not read); Deck 1.1’s Holmström frame.

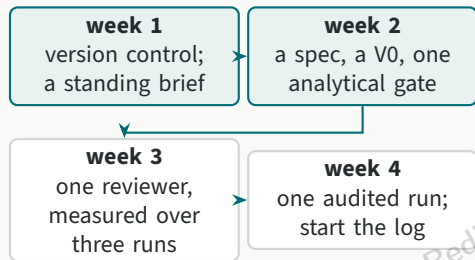
Five Applications, Read as Team-Design Questions

application, per the abstract	the team-design question it becomes
(i) sandbagging: a more capable agent pretends to be less capable	the sub-agent that reports “not feasible”: hiding capacity, or lacking it? Probe it on a task you know it can do
(ii) an alignment–interpretability trade-off: substitutes in the instrument, complements in value	danger (ii): a reviewer whose reasoning you can read is worth more than a stronger one you cannot
(iii) discipline via peer scoring	the verifier sub-agent (Deck 6.3): a second agent scores the first, on a rubric written first
(iv) coupling rewards to induce competition among agents	two solvers on one gate, one comparison; neither sees the other’s answer
(v) scalable oversight and reward shaping	the gate itself — and what a loop does once it is told what will be measured (Deck 1.1’s Lucas critique)

► The five stylised examples are Deck 6.3’s instruments, stated as a theory.

Source: Bergemann, Koh & Morris (2026), arXiv:2609.01595 v1, 1 Sep 2026, abstract; the right column is this course’s reading; Deck 1.1’s Holmström and Lucas-critique frames.

Your First Thirty Days, and the Job Description



the furthest point you can still
check is the furthest you should go

- ▶ Four weeks, one habit each, on a project you already have. Read first: Korinek (2025), NBER 34202, the economist's own manual, an agent built step by step; Galiani, López & Sosa (2026), NBER 35588, what agency bought on econometric tasks.
- ▶ You remain the **principal investigator**, and the **project manager** of Deck 6.1: gate every good, assign it, cross-verify it.
- ▶ **Agents lay bricks quickly. Remain the client of a prior other than your own.**

Source: Korinek (2025), NBER WP 34202; Galiani, López & Sosa (2026), NBER WP 35588; Parasuraman & Manzey, *Human Factors* (2010); Miao & Feng, *AI for Economists*, ch. 15, §5.

What the Course Promised, and Where It Was Built

promised in Deck 1.1

text measures and a small transformer

a grounded agent that leaves a trace

citations verified

numerical outputs verified

failure paths verified

reproduction from a clean, offline start

discharged

Session 5's lab, end to end

block A: 7 readable turns

block D: each sentence carries its meeting

block B: 96 numbers recomputed

block D's three failures; block F's gate

fixed seeds, no network, hashes checked

- ▶ The course opened on a sentence: fluent output is not a result — a result survives a check it could have failed. Block B is that sentence with numbers in it.
- ▶ **So demand four things of anything an agent hands you: the trace that made it, the numbers recomputed from the source, the claim stated apart from the arithmetic, and the rule that would have stopped it. Short of all four, it is a draft.**

Source: Deck 1.1, what students build and verify; this session's lab, blocks A to F.